PORT OF SEATTLE

STORMWATER TREATMENT BMP EVALUATION

Prepared for:
Port of Seattle
Pier 66
Seattle, WA 98111

Prepared by:
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In association with:
AGI Technologies
Am Test, Inc.
Resource Planning Associates

MARCH 1997

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TABLE OF CONTENTS

Page

Executive Summary i

Introduction 1
- Previous Research 1
- Sweeping Technologies 2

Data Collection Program 4
- Mechanical Analyses 5
- Chemical Analyses 5

SIMPTM Overview 6
- Washoff and Accumulation 6
- Sweeper Pickup Performance 7
- Sediment Trapping Catchbasins 8

SIMPTM Calibration 9

Annual TSS Load Reduction from BMP's 10
- Average Rainfall Year Development 10
- TSS Loadings 11
- Dry Versus Damp Pavement Sweeping 11

Annual Pollutant Loads and Concentrations 12

Study Conclusions 13

References 15

PORT OF SEATTLE
STORMWATER TREATMENT BMP EVALUATION

EXECUTIVE SUMMARY

Introduction

In May, 1996, the Port of Seattle contracted with Kurahashi & Associates, Inc. (KAI) to conduct an evaluation of selected stormwater treatment best management practices (BMP's) for the Port's marine terminals. The study's objective was to evaluate the stormwater pollutant removal effectiveness of new high efficiency pavement sweepers in combination with conventional sediment trapping catchbasins and to determine if this combination technology is essentially equivalent to the pollutant removal efficiencies of wet vaults. For new marine facilities, the only stormwater treatment BMP that is technically feasible and currently approved by the Department of Ecology is the wet vault.

The type of Port facilities for which sweeping and sediment trapping catchbasins were evaluated is the cargo container yard. The Port currently owns five such facilities covering a total area of approximately 400 acres. The Port is currently undertaking major expansion projects at two of the five yards. Hence, the potential cost effectiveness of high efficiency sweeping is most applicable to these types of facilities. The immediate project of interest is the Southwest Harbor Project. This project, currently under construction, will expand an existing container yard by 250 acres.

The Port was motivated to evaluate sweeping because of the results of recent studies conducted by KAI (Sutherland and Jelen, 1996b and 1997). These recent studies have concluded that high efficiency sweepers and/or sweeping operations can provide significant reductions in various pollutants being transported in stormwater. These conclusions are in sharp contrast to the conclusions reached in 1983 by the USEPA sponsored Nationwide Urban Runoff Program (NURP).

The NURP investigated urban stormwater quality over a three year period at approximately 28 locations throughout the United States. The NURP's studies of street sweeping concluded that the practice was largely ineffective at reducing the event mean concentration (EMC) of pollutants in urban runoff. This conclusion was reached mainly because the street sweepers that were tested were not able to effectively pick up very fine accumulated sediments that can often be highly contaminated with other pollutants. These same sediments that have accumulated on paved areas that are directly connected to a stormwater inlet and sewer system have been identified over and over again as the primary source of urban nonpoint pollutants entering the receiving waters of the nation.

Study Overview

The stormwater quality control study that this report will describe involved the monitoring of accumulated sediments and their associated pollutants on various activity areas of an existing Port container storage yard. This pollutant accumulation data were obtained by hand sweeping nine, 2,000 square foot test sites on an existing operational container yard over a two month period. These test sites differed by the type of yard activity. This information, along with the physical and chemical characteristics of the sampled sediment and observed rainfall at a nearby site was used to calibrate a stormwater quality computer model for each of these activity areas.

Using the calibrated model parameters from each of these activity areas, the model was used to simulate the sediment and associated pollutant accumulation and washoff that will occur over an average year from a representative 10-acre container yard. The average representative precipitation year was obtained from a detailed statistical analysis of the entire 30 year precipitation record at Sea-Tac International Airport. The analysis resulted in identifying and piecing together the most average months in the record (based on an analysis of seven different parameters) to yield the most average year. The model simulations dealt with alternative frequencies of sweeping (e.g. daily to monthly) and catchbasin cleaning (monthly to annually), along with alternative sizes of catchbasins (e.g. normal, at 10 cubic feet per 15,000 ft^2 of area; or enlarged, at 37.8 cubic feet for the same area)

The results of these model simulations, which will be briefly presented in the next section, were then compared to the results obtained from the simulated performance of a wet vault designed to the Puget Sound water quality criteria. The sediment and associated pollutant removals of the wet vault were based on a set of modified Stoke's Law equations needed to compute settling velocities for various sizes of transported sediments. These calculations were processed in a spreadsheet using outputs from the computer model as the starting point. These model simulations assumed no sweeping but they did assume that enlarged catchbasins would be used and would be cleaned at the same frequency that the comparable sweeping with no wet vault runs had assumed. By holding size and cleaning frequency of the catchbasins the same, the results will truly show the anticipated pollutant reductions resulting from alternative levels of sweeping.

As stated earlier the computer simulation considered the effect of sweeping frequency on the reduction of pollutants in individual storms that occur throughout an average representative precipitation year. It also considered the effects of rainfall periods on both the ability to sweep and the efficiency of the sweeper. Rainfall periods are of three types: dry period (high sweeping efficiency), rainy periods that produce runoff (sweeping cannot occur), and rain periods that cause damp pavement conditions without runoff (sweeping occurs but at reduced pick-up efficiency).

Unfortunately, the computer model does not have the ability to alter the basic performance characteristics of the sweeper during a simulation. So, two sets of model simulations were obtained: one assumed the performance characteristics of damp pavement sweeping and the other assumed those of dry sweeping.

It should be noted that this study used the SIMPTM (Sutherland and Jelen, 1993) stormwater quality model. The SIMplified Particulate Transport Model (SIMPTM) is a continuous stormwater quality model that can accurately simulate the stormwater pollutant loadings and expected load reductions from best management practices (BMP's) such as street sweeping and using and cleaning water quality inlet and manholes. SIMPTM is unique in its ability to simulate the accumulation, washoff and BMP removal of sediment and its associated pollutants (Sutherland and Jelen, 1996a). Other applications of the model have demonstrated its ability to accurately simulate the observed pollutant loads and concentrations from gauged urban basins such as those monitored for the City of Portland's NPDES stormwater permit and the City of Bellevue's NURP.

SIMPTM model inputs that characterized the various sweeping operations were based on previous work completed by KAI in the area of sweeper performance testing. KAI has conducted detailed performance tests of eight different types of sweeping equipment, using two different operating speeds and twelve sites located throughout four states in the country. Without this information, and the ability of SIMPTM to utilize it to accurately simulate the interaction of sweeper pick-up with sediment accumulation and washoff, the following results could not have been obtained.

Study Results

The results of the study show that high efficiency sweeping will provide significant reductions in annual pollutant loadings of the five pollutants that were simulated which included total suspended solids (TSS), total copper (Cu), total phosphorus (T-P), total lead (T-Pb), and total zinc (T-Zn). These reductions range from an estimated 29% to 78% depending on the sweeping frequency, the catchbasin size and the specified pollutant.

When compared to the estimated pollutant removal of wet vaults (in combination with enlarged catchbasins), sweeping (in combination with normal or enlarged catchbasins) is projected to equal or exceed the removals obtained for total copper, total phosphorus and total zinc. However, wet vaults (in combination with enlarged catchbasins) are projected to provide a higher level of pollutant reduction for TSS and total lead at all simulated sweeping frequencies.

A brief sample of the model results are presented in Table 1. A detailed presentation of the extensive model results will follow later in this report. As mentioned earlier in the text, the model does not have the capability of varying the pick-up performance characteristics of the simulated sweeping operation during the year long simulation. For the damp pavement or dry pavement conditions noted in Table 1, this means that when sweeping is scheduled to occur and provided that runoff is not occurring at that time (i.e. sweeping would not be simulated if runoff conditions exists), SIMPTM assumes the pick-up performance characteristics of damp pavement or dry pavement, respectively, throughout the entire annual simulation. Since damp pavement conditions will occur on occasion, throughout the year along with dry pavement conditions, one can probably assume that the actual pollutant removal effectiveness from the model simulation would be somewhere in between the two removals shown for each pollutant listed.

Please note that the modeled difference between these two sweeping conditions (i.e. damp or dry pavement) is not very significant. This is due primarily to the overland washoff hydraulic characteristics which are very sensitive to the sweepers removal of coarse particles (i.e. 250 microns or greater) which is quite good in both cases. Since both sweeping operations (i.e. damp versus dry) are also considered quite effective in the pick-up of mid range particles (i.e. 63 to 250 microns), the net effects of the greater pick-up performance of the fine material (i.e. less than 63 microns) is not that great.

Dramatic changes in performance may occur between damp and dry sweeping when occasional concentrated runoff conditions like gutter flow or swale flow are present on the site. It is important that the Port uses for any field testing or program implementation the most efficient pavement sweeper that exists which is the EnviroWhirl dry sweeper manufactured by EnviroWhirl Technologies, Inc. out of Centralia, Illinois.

The following study conclusions are offered.

Study Conclusions

1. For marine cargo handling and storage facilities, the pollutant removals associated with high efficiency sweeping at a weekly frequency in combination with normal catchbasin inlets, cleaned annually or semi-annually, is essentially equivalent to those removals obtained by wet vaults.

2. Sweeping appears to be competitive with wet vaults even though a sweeper is unable to access all of the container yard. A sweeper is unable to sweep all of the site because of the presence of parked container trailers which cannot be moved to accommodate the needs of sweeping. Despite this limitation, sweepers may be essentially equivalent to wet vaults because of their high pick up efficiency, as well as their ability to remove dissolved pollutants.

3. Wet vaults are able to remove only sediment and the associated particulate pollutants. They are not able to remove dissolved pollutants. In contrast, high efficiency sweepers are able to remove dissolved pollutants since this fraction is contained in the accumulated sediment and can be removed by the sweeper during dry periods.

4. TSS is the characteristic of stormwater that is commonly used to compare the performance of alternative stormwater BMP's. Sweeping may be unable to achieve the high TSS removal efficiency simulated for wet vaults.

5. High efficiency sweeping may be more efficient than wet vaults in the removal of pollutants that tend to have a significant fraction that is dissolved such as copper, zinc, and phosphorus.

6. Wet vaults may be more efficient than high efficiency sweeping in the removal of pollutants that tend to have a significant fraction in the particulate state such as lead.

7. For high efficiency sweeping to be used as an alternative to wet vaults. the sweepable portions of the site must be swept at least weekly. For the Southwest Harbor Project mentioned earlier, this is anticipated to require the full-time dedication of at least one sweeper. The model simulations assumed that no sweeping will occur when runoff conditions exist. This is the actual practice that needs to be implemented in the field.

8. The efficiency of wet vaults in the study is calculated number based on a modified Stoke's Law equations needed to compute settling velocities for various size of transported sediments. It is probable that wet vault efficiency is actually lower in practice. As a result, high efficiency sweeping may be considered an even more viable solution under actual field conditions.

Given the potential uncertainty of the model simulations in Table 1, ranges of possible removal efficiencies are presented in Table 2. Also presented in Table 2 is the effect of sweeping frequency. With regard to sweeping frequency, the model suggests there may be significant benefit in sweeping weekly as opposed to biweekly. However, there may be little incremental benefit of sweeping twice weekly in comparison to weekly.

With regard to overall removal efficiencies, the model simulations presented in Table 1, include two assumptions that are altered to provide the range in Table 2. First, the model assumes that the sweeper removes the same fractions of dissolved pollutants associated with particulate material during damp pavement conditions as it does during dry pavement conditions. This may not be the case since some of the dissolved fraction could be transported to the catchbasins before it can be swept. Secondly, the model assumes that material that accumulates under the trailers is not washed from the site since the topography of a container yard is such that stormwater typically does not flow beneath trailers except in extreme events. Regardless, material is moved from under the trailers into the alleyway. It is likely, however. that because of the overland flow condition, much of the material that is washed from beneath the trailers does not reach the catchbasins; rather, it settles on the pavement and could be removed by sweeping immediately after the event. Nonetheless, the low end of each range presented in Table 2 assumes that none of the dissolved pollutants are captured by sweeping during the damp pavement conditions sweeping and

that the parked trailers do not block the potential transport of material beneath trailers so it is assumed to wash directly to the catchbasins. Hence, the vault is given the opportunity to capture this material whereas the sweeping is not.

With regard to wet vaults, SIMPTM predicts a removal efficiency of about 90% for TSS (Table 1). This is greater than other predictive methods such as USEPA (1986). The USEPA methodology estimates removal efficiency using rainfall statistics and settling velocity data generated in the NURP study. Based on the concept of the volume specified by the Department of Ecology (1992), the estimated performance according to USEPA (1986) is about 75%. The ranges in the removal efficiencies for the wet vault in lower end of the ranges shown in Table 2 reflect somewhat more conservative assumptions as described in the report. What Table 2 still indicates is that ranges for sweeping are similar to the range for wet vaults for those pollutants that tend to be more dissolved.

Table 1

ANNUAL ALTERNATIVE POLLUTANT LOAD REDUCTIONS FROM SIMPTM SIMULATIONS

	Weekly Sweeping with Enlarged Catchbasins		Wet Vault with Enlarged Catchbasins
Pollutant	Damp Pavement	Dry Pavement	No Sweeping
TSS	65%	68%	91%
Total Phosphorus	52%	55%	46%
Total Lead	55%	60%	81%
Total Zinc	47%	52%	45%
Total Copper	53%	56%	46%

Enlarged catchbasins are cleaned twice during the simulation.

Table 2

EXPECTED RANGE OF ANNUAL POLLUTANT LOAD REDUCTIONS ASSOCIATED WITH VARIOUS SWEEPING FREQUENCIES

	Twice Weekly Sweeping	Weekly Sweeping	Biweekly Sweeping	Wet Vaults
TSS	45 - 70%	45 - 65%	40 - 60%	75 - 90%
Total Phosphorus	35 - 60%	30 - 55%	20 - 40%	35 - 45%
Total Lead	40 - 60%	35 - 60%	30 - 50%	65 - 80%
Total Zinc	30 - 55%	25 - 50%	20 - 40%	35 - 45%
Total Copper	35 - 60%	30 - 55%	20 - 40%	35 - 45%

Normally sized catchbasins are cleaned once during the year.

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INTRODUCTION

In May, 1996, the Port of Seattle contracted with Kurahashi & Associates, Inc. (KAI) to conduct an evaluation of selected stormwater treatment best management practices (BMP's) for the Port's marine terminals. The study's objective was to evaluate the stormwater pollutant removal effectiveness of new high efficiency pavement sweepers in combination with conventional sediment trapping catchbasins and to determine if this combination technology is essentially equivalent to the pollutant removal efficiencies of wet vaults. For new marine facilities, the only stormwater treatment BMP that is technically feasible and currently approved by the Department of Ecology is the wet vault.

The type of Port facilities for which sweeping and sediment trapping catchbasins were evaluated is the cargo container yard. The Port currently owns five such facilities covering a total area of approximately 400 acres. The Port is currently undertaking major expansion projects at two of the five yards.

The immediate project of interest is the Southwest Harbor Project. This project, currently under construction, will expand an existing container yard by 250 acres. Wet vaults are estimated to have a life cycle cost of about $18 million for this expansion project due primarily to the structural requirements of the vault covers that need to fully support large, heavy container transport vehicles called "top picks". In contrast, high efficiency weekly pavement sweeping with normal sediment trapping stormwater inlets cleaned annually will have a life cycle cost of approximately $2 million for the same expansion project.

The Port was motivated to evaluate sweeping because of the results of recent studies conducted by KAI (Sutherland and Jelen, 1996b and 1997). These recent studies have concluded that high efficiency sweepers and/or sweeping operations can provide significant reductions in various pollutants being transported in stormwater. These conclusions are in sharp contrast to the conclusions reached in 1983 by the USEPA sponsored Nationwide Urban Runoff Program (NURP).

Previous Research

The Nationwide Urban Runoff Program (NURP) studies of street sweeping effects on stormwater quality (USEPA 1983) concluded that street sweeping was largely ineffective at reducing the event mean concentration (EMC) of pollutants in urban runoff. This conclusion was reached mainly because the street sweepers tested were not able to effectively pick up very fine accumulated sediments that can often be highly contaminated.

In general, street sweeping equipment of the NURP era was unable to effectively pick up the very fine, highly contaminated, sediment that accumulate on impervious areas such as streets, driveways and parking lots. These same sediments, located on paved areas that are directly connected to a city's storm drainage system, have been identified over and over again as the primary source of urban nonpoint pollutants entering the receiving waters of the United States.

Broom sweepers of that era removed litter and large dirt particles well, but contaminants are known to concentrate primarily in the fine particle sizes (e.g. less than 63 microns). However, these finer and much more pollutant-laden particles were largely left behind, and moreover, they were left exposed to be even more readily entrained in washoff since their armoring shelter by larger sediment particles was removed.

However, recent studies by KAI staff over a period of four years show clearly that the NURP conclusions from the early 80's are no longer valid today. This is largely because of the considerable increase in street sweeping's effectiveness at removing the smallest particles. Examples of this improvement include the following:

1. Even most mechanical sweepers (i.e. broom and conveyor belt) now available are much more effective at picking up fine sediments.

2. Tandem sweeping operations (i.e. mechanical sweeping followed immediately by a vacuum-assisted machine) have been found to be even more effective at fine sediment pickup.

3. Regenerative air sweepers have been refined considerably since their infancy during the NURP era, and have also been found to be effective at fine sediment pickup.

4. A revolutionary new vacuum-assisted dry sweeper called the Enviro Whirl has greatly advanced the technology of fine sediment pickup and containment.

Sweeping Technologies

The pickup performance for the NURP era sweepers show typical values based on a previous analysis of the Bellevue, Washington NURP data (Sutherland, 1990), as summarized in Pitt, 1985. Having been a consultant to the City of Bellevue during the NURP study, Sutherland had direct access to the street sweeper pickup performance data collected as part of that study. The sweeper tested at that time was a Mobil standard mechanical broom street sweeper, probably manufactured around 1978. It provides the baseline against which several modern street sweeping technologies are compared for immediate pickup rate and expected long-term washoff load reduction.

Against this, the performance of a newer mechanical (i.e. broom and conveyor) sweeper was compared, in order to establish the level of improvement achieved in types of sweepers still in wide use. Data for this comparison was obtained when Sutherland and Jelen measured the pickup performance of a newer mechanical sweeper which was a 1988 Mobil, as part of a Portland study (HDR, 1993).

Research by Sutherland and Jelen (1996b and 1997) has identified three promising technologies that may provide significant improvements in performance beyond that observed for NURP era or mechanical sweepers. For each, the sediment pickup from sweepings by each technology was measured in the field under a variety of conditions. Resulting removals were obtained for each of eight particle size ranges. These show significantly greater removals for each of these new technologies than those typical for sweepers from the early 1980's.

The first technology is the use of a tandem sweeping operation. A tandem operation involves two successive cleaning passes, first by a mechanical (i.e. broom and conveyor belt) sweeper, then immediately followed by a vacuum-assisted sweeper. The pickup performance of a tandem operation using the Mobil

broom sweeper followed by a TYMCO vacuum sweeper was monitored for over a year in a medium-density residential area located in Southeast Portland, Oregon. The detailed description of this study and its results can be found in HDR, 1993, and were briefly summarized in Alter, 1995.

The second technology is the stand-alone use of a regenerative air sweeper. Regenerative air sweepers blow air onto the pavement and immediately vacuum it back in order to entrain and filter out accumulated sediments. Regenerative air machines were just in their infancy during the NURP era, and were not extensively tested at any of the NURP sites. Regenerative air sweepers are generally considered to be good at removing fine sediment, if the accumulated loading is not too great. Sutherland and Jelen (1995) measured the pickup performance of the Elgin Crosswind regenerative air sweeper in and near SeaTac International Airport on April 21, 1995.

The third technology is the stand-alone use of a new, highly effective, vacuum-assisted dry sweeper called the Enviro Whirl I developed and manufactured by Enviro Whirl Technologies Inc., located in Centralia, Illinois. This sweeper applies technology developed and still used to remove spilled coal and coal dust from along railroad tracks. The technology has also been applied to clean similar materials from industrial sites where complete removal without leakage of airborne particles is important.

From these demands have evolved a technology that is extremely efficient at removing the finest particles and preventing their escape into the air. In contrast, most other units, especially mechanical types, trail a visible cloud of dust behind in the air and on the street.

The Enviro Whirl I combines the important elements of tandem sweeping into a single unit. It uses rotating sweeper brooms within the powerful vacuum head to provide both mechanical and aerodynamic particulate removal. Its' unique filtration system removes particles greater than 2 microns from its' air emissions. Data used to compare the sweeping performance of this technology to others was obtained by Sutherland and Jelen in an April 24, 1995 test prepared by the City of Las Vegas, Nevada (during an air quality conference) and in Centralia, Illinois during September, 1995.

This data reveals marked improvements in the street sweeping technology that results in much more effective pickup of accumulated sediments. Using the NURP-era broom sweepers as a baseline, performances were compared for improved mechanical sweepers and promising sweeping technologies (Sutherland and Jelen, 1997). As a result, it becomes clear that street sweeping is now capable of removing significant pollutant loads from urban surfaces and effecting significant reductions in urban pollutant washoff.

The stormwater quality control study that this report will describe involved the monitoring of accumulated sediments and their associated pollutants on various activity areas of an existing Port container storage yard. This pollutant accumulation data were obtained by hand sweeping nine, 2,000 square foot test sites on an existing operational container yard over a two month period. These test sites differed by the type of yard activity. This information, along with the physical and chemical characteristics of the sampled sediment and observed rainfall at a nearby site was used to calibrate a stormwater quality computer model for each of these activity areas.

DATA COLLECTION PROGRAM

The general approach to the study was to obtain data that described the rate of pollutant accumulation on an existing container yard. The collected data would then be used to calibrate a computer model that can simulate the washoff of sediment and its associated pollutants and the effect of the various alternative control technologies on the actual pollutant loadings.

The Port of Seattle's T-5 Yard was selected for the data collection program. The T-5 Yard was visited by the project team in late May, 1996. The project team consisted of Port of Seattle staff, KAI staff and Resource Planning Associates (RPA) staff who were contracted to collect the accumulation data over a two month period. The purpose of the visit was to select accumulation test areas that were representative of the various on-going yard activities.

Nine test sites were identified within three different yard activity areas with three test sites for each activity area. The activity areas were the alley ways between stored containers, the alley ways between parked trailers and the area beneath the trailers themselves. Each of the test areas were 2000 square feet in size.

The data collection program was designed as follows:

1. Three test sites per yard activity area.

2. The yard activity areas were named Container Alleys, Trailer Alleys and Trailer Parking.

3. All test sites were hand swept and vacuumed at the start of the two month test period.

4. One of the test sites in each of the yard activity areas was designated as the "Week 1" site which means they were cleaned or tested every seven days or weekly.

5. Another one of the test sites in each of the yard activity areas was designated as the "Week 2" site which means they were cleaned or tested every two weeks.

6. The last of the test sites in each of the yard activity areas was designated as the "Week 4" site which means they were cleaned or tested every four weeks or monthly.

The program was designed to run over a two month period. Forty-five samples would have been collected over this period provided that dry pavement conditions existed on the designated sampling days which means no rainfall interference. Dry pavement conditions are needed to conduct the hand sweeping and mechanical vacuuming procedures needed to collect the samples.

The program began on May 31, 1996, and ran through July 30, 1996. Thirty-nine samples were collected. No measurable rainfall occurred for the first thirteen days of the program. Rainfall interference in mid to late July reduced the number of "Week 1" samples taken at each activity area to six instead of the eight that were initially planned.

A rainfall gauge was initially installed at the T-5 Yard. However, it was destroyed a few weeks into the data collection program. Hourly rainfall data collected at nearby SeaTac International Airport was used for the model calibration.

Mechanical Analyses

AGI Technologies of Portland, Oregon provided the mechanical analyses of the yard dirt samples under contract to KAI. All work was performed in general accordance with the appropriate ASTM test methods except where procedures were modified as required by the Scope of Work. To assure continuity, all sieving and subsequent weighing was performed by the same individual on AGI's staff.

Mechanical grain size analysis was performed on 39 samples provided by KAI. The following information was recorded:

1. Wet weight of sample.
2. Dry weight of sample (after drying to constant weight at a temperature of 120 degrees F).
3. Percent moisture.
4. Dry weight and percentage retained for each of the following eight particle size groups:

    a. less than 63 microns (No. 230)
    b. 63-125 microns (No. 120)
    c. 125-250 microns (No. 60)
    d. 250-600 microns (No. 30)
    e. 600-1000 microns (No. 18)
    f. 1000-2000 microns (No. 10)
    g. 2000-6370 micron (1/4 in.)
    h. >6370 microns (>1/4 in.)

The dry weights for each sieve fraction was totaled and compared to the sample net dry weight. Weight gain (loss) was noted. All testing was accomplished with stainless steel equipment provided by KAI. All weights were measured in metric units to two significant figures (nearest 0.01 gram). The results of the mechanical analyses are presented in Table 1. (All the tables are included in order following the text of the report.)

Chemical Analyses

After the mechanical analysis, samples were composited back into three fractions in labeled sealed plastic bags for chemical analyses by others. The composite samples were constructed and labeled as shown: 1) <63 microns, "F" for Fine, 2) 63 to 250 microns, "M" for Medium, and, 3) 250 to 6370 microns, "C" for Coarse. The portion greater than 6370 microns was discarded.

AMTEST, Inc. of Beaverton, Oregon provided the chemical analyses of the composite yard dirt samples under contract to KAI. All work was performed in general accordance with the appropriate EPA test methods.

The three composited fractions (i.e. F, M and C) of four separate samples taken from each of the three activity areas were tested for ICP-Metals (EPA 6010) and Total Petroleum Hydrocarbons (TPH) (EPA 418.1M). The initial samples collected on May 31, 1996 were tested along with the first "Week 1", "Week 2" and "Week 4" samples collected on June 5, 1996, June 13 and 14, 1996 and June 27, 1996, respectively. The results are shown in Table 2. Although the ICP-Metals analysis provided results on thirty-three different metals, Table 2 only includes the results for copper, lead and zinc.

The three composited fractions of two separate samples taken from each of the three activity areas were also tested for Chemical Oxygen Demand (COD) (EPA 410.4), Ortho Phosphorus (EPA 365.3) and Total Kjeldahl Nitrogen (TKN) (EPA 351.3). The initial samples collected on May 31, 1996 were tested along with the first "Week 4" samples collected on June 27, 1996. The results are shown in Table 3.

Several months after the model study described in the next section was underway, it was determined that some additional chemical analyses was needed. The topic of interest was the occurrence and magnitude of dissolved metals and Total versus Ortho Phosphorus in composited fractions (i.e. F, M and C) of samples obtained from each of the activity areas.

To determine the magnitude of dissolved pollutants within specific particle size ranges of accumulated yard dirt, TCLP Metals (EPA 1311) tests were run for copper, lead and zinc for the composited fractions of several samples from each of the activity areas.

The results are presented in Table 4. It is interesting to note that dissolved metals were found to exist in all three of the yard dirt fractions. In fact, the medium fraction (i.e. 63 to 250 microns) had dissolved metals concentrations from leachate that equaled or exceeded those observed from the fine fraction (i.e. less than 63 microns). This data clearly suggests that if significant amounts of fine, medium and coarse yard dirt could be removed by a sweeper in the dry state before rainfall interacts with it, dissolved as well as particulate metals will be removed.

This observation is quite significant since wet vaults are essentially ineffective at removing dissolved pollutants. Whereas, high efficiency sweeping may be able to significantly reduce the magnitude of dissolved pollutants like metals found in stormwater. Similar observations could be made for Ortho Phosphorus (EPA 365.3) especially when it is compared to Total Phosphorus (EPA 365.2) as shown in Table 5.

SIMPTM OVERVIEW

It should be noted that this study used the SIMPTM (Sutherland and Jelen, 1993) stormwater quality model. The SIMplified Particulate Transport Model (SIMPTM) is a continuous stormwater quality model that can accurately simulate the stormwater pollutant loadings and expected load reductions from best management practices (BMP's) such as street sweeping and using and cleaning sediment trapping catchbasins and manholes. SIMPTM is unique in its ability to simulate the accumulation, washoff and BMP removal of sediment and its associated pollutants (Sutherland and Jelen, 1996a). Other applications of the model have demonstrated its ability to accurately simulate the observe pollutant loads and concentrations from gauged urban basins such as those monitored for the City of Portland's NPDES stormwater permit and the City of Bellevue's NURP.

Washoff and Accumulation

SIMPTM is a continuous, self-contained stormwater quality model. It divides hourly precipitation records into rainfall events and provides monthly and annual statistics for later analysis. For each event, it forms a runoff hydrograph, used to continually simulate sediment and bound pollutant transport using the Yalin-Einstein and Foster-Meyer equations, to simulate the capacity of the hydrograph to transport available accumulated sediment from paved areas. This has been shown by Ellis and Sutherland (1979)

to be much more accurate than the empirically-based exponential washoff used, for example, in EPA's Storm Water Management Model (SWMM).

The model also accounts for sediment deposition, armoring, and resuspension processes. Between events, SIMPTM calculates dry deposition and resuspension processes and models scheduled cleaning of streets, parking lots, catchbasins, or maintenance hatches. Overall removals from these practices are provided by SIMPTM based upon measurable data, rather than input by the user as most stormwater quality models require. Any excess erosion remains available for further simulation, so that actual accumulations may often exceed the *equilibrium* load previously assumed by many to be a *maximum* limit to accumulation.

SIMPTM has been used extensively in the Portland, Oregon area as part of surface water management and water quality management plan development since 1988. It has been used in the Seattle area several times and its predecessor in Anchorage, Alaska, Reno/Sparks, Nevada, and throughout urban Illinois. The United States Geological Survey (USGS) recently applied the model in the Dallas, Texas area and concluded that it provided the most accurate simulation results when compared to several other models that were tested including SWMM (USGS, 1992). For a more detailed description of SIMPTM and a discussion of how it was recently calibrated to an extensive NPDES data set collected in Portland, Oregon, please refer to Sutherland and Jelen (1996a).

Sweeper Pickup Performance

The ability of high efficiency sweeping to reduce overall pollutant washoff loads depends on several things. First is the sweeper's innate ability to remove accumulated sediment. Another is the environmental dynamics of sediment accumulation and resuspension, and of sediment washoff during storm events plus suspended sediment removal by downstream water quality controls.

The SIMPTM program can accurately simulate this complicated interaction of accumulation, washoff, and street sweeper pickup that occurs over a period of time (Sutherland and Jelen, 1993). The street sweeping component of the SIMPTM model was based on the results of Pitt's street sweeping study conducted in San Jose, California (Pitt, 1979). This model was confirmed in additional studies conducted in Alameda County, California (Pitt and Shawley, 1982) and in Washoe County, Nevada (Pitt and Sutherland, 1982).

These studies found that sweeping removes little, if any material below a certain base residual which was found to vary by particle size. Above that base residual, the street sweepers removal effectiveness was described as a straight line percentage which varied by particle size.

Figure 1 illustrates the street cleaning component and equations used by SIMPTM. (All figures are included in order following both the text and tables.) For each of eight size groups, the amount removed (*Prem*) is proportional to the initial accumulation (*Po*) in excess of a base residual (*SSmin*) by a sweeping efficiency (*SSeff*):

$$Prem = SSeff * (Po - SSmin) \text{ for } Po > SSmin$$

Therefore, to describe a unique street sweeping operation one simply needs to know the operations SSmin and SSeff values for each of the eight particle size ranges simulated by SIMPTM. Note that SSeff is dimensionless, while that for SSmin must match that for accumulation, usually either pounds per curb

mile or pounds per paved acre.

Figure 2 shows an example of how this model component actually compares to real pickup performance data for each of the eight particle size groups. The plotted points are the data obtained from monitoring the tandem street sweeping operation on Portland's Sellwood drainage basin (HDR, 1993). Note that the correlation coefficients (R^2) for the fits of the eight particle size groups ranged from 94.3% to 99.9%, so the model is doing an excellent job of reproducing the actual observations. These high explained variations (i.e. high R^2) were typical of all of the model fits to the pickup data from the various sweeping technologies.

Table 6 compares the SSmin parameters calibrated to model each of the five sweeping technologies. It shows dramatic improvements in reducing residual loadings for all the newer technologies when compared to the NURP sweepers. While both tandem sweeping and the Elgin Crosswind regenerative air are very impressive, the across-the-board zero residual loadings for the Enviro Whirl I is the best possible.

Table 7 compares the corresponding marginal sweeping rate, SSeff, when sweeping loads exceeding the threshold SSmin. They were also calibrated to model each of the five sweeping technologies. The results mirror those for the SSmin parameter, and show impressive removal efficiencies above the residential loadings. Dramatic improvements are again evident since the NURP era. It must be recognized that this table shows only a marginal removal rates. The overall removals must also incorporate the residual loading that always remains after sweeping. Thus, although the rates of the Elgin Crosswind (regenerative air) and the Enviro Whirl I for the finer particle size groups may not look that impressive, their residual loadings are very low, or even zero, resulting in overall removal efficiencies that are essentially the same as the rate shown. Other technologies with larger SSmin's would be significantly less efficient.

Sediment Trapping Catchbasins

Very little research has been conducted on the sediment retention effectiveness of sediment trapping inlets or catchbasins. However, experiments by Lager et al. (1977) concluded that sediment accumulation in catchbasins is a function of the incoming sediment sizes, the catchbasin volume available to trap sediment, and the runoff flow rate entering the catchbasin. As expected, the larger particles had the highest retention percentage.

An initially clean catchbasin would retain approximately 45 percent of the incoming sediment until it became about half full. The efficiency then quickly drops to 0 as the trap fills to 60 percent. This critical point was defined as the breakthrough point. SIMPTM uses log-log regressions to relate the capture rate for each group to the retention rate (X), or flow over available trap storage exponentially:

$$Capfrac = 1/2(1 - A_{trap}X^{Btrap})$$

The two parameter set A_{trap} and B_{trap} are coded into the program. Each set of eight size-group values (i.e. A_{trap} and B_{trap}) can be redefined. The default parameters for A_{trap} and B_{trap} currently used by the program were based on a calibration of the SIMPTM model to the extensive Bellevue NURP data set (Sutherland, 1991).

SIMPTM CALIBRATION

As noted in the previous section, the SIMPTM program has the ability to simulate the accumulation and washoff of sediments and their associated pollutants. The model calibration process generally involves the adjustments of parameter values to reproduce observed runoff volumes and pollutant loads. However, since no end-of-pipe stormwater flow and pollutant concentration data was obtained, the calibration focused on reproducing the observed sediment accumulations for each of the activity areas during the two month sampling period.

Using the hourly rainfall data observed at the SeaTac International Airport during the two month sampling period, approximately eight runoff producing rainfall events were identified. Runoff producing rainfall events were those events that satisfied one of the three minimum depth versus time criteria:

1. 0.05 inches in one hour
2. 0.08 inches in three hours, or
3. 0.10 inches in six hours.

SIMPTM modeled rainfall losses using a single-parameter exponential curve that varied with rainfall depth. Figure 3 illustrates this curve. Once the maximum loss was specified, the exponential rate was automatically set so that the loss matched rainfall at the start of an event. This rainfall loss, which depends mainly on pavement texture or condition, was set to the value recently calibrated from Portland, Oregon runoff data — 0.05 inches.

Most SIMPTM parameters involving the hydraulics of washoff and sediment entrainment were related to measurable or fairly standard quantities, and were set as part of the activity area characterization.

SIMPTM modeled accumulation as a constant deposition rate (mass per day) that became balanced over time by proportional resuspension rate due to wind and traffic (per day). As a result, accumulation approached an equilibrium limit determined by the deposition rate divided by the resuspension rate. Figure 4 illustrates the two-parameter exponential function used. One parameter specified the equilibrium accumulation limit; the other, the resuspension rate. Within each activity category, each size group shared a constant rate, but apportioned the limit using a specified average particle size distribution that was obtained directly by sieving the accumulation samples which was presented in an earlier section.

The actual sampling event was simulated by the model to be a "perfect" sweeping event in which the minimum residuals (SSmin) are zero and the sweeper pickup efficiencies (SSeff) were all 100%. A "perfect" sweeping event was simulated by SIMPTM on each of the days in which an accumulation sample was obtained.

With all the washoff parameters set and all but one of the accumulation parameters set to reasonable values observed during other calibrations, the accumulation rate was varied for each of the three activity areas until one uniform rate was found to provide the best overall match. The best overall match was determined by visually examining the model's simulated sediment accumulation values (lbs/acre) against the actual sample weight that was obtained. Each activity area had three test sites that represented the three allowable accumulation periods of 1 week, 2 weeks and 4 weeks, described earlier.

The results of the calibration are presented in Figures 5, 6, and 7 which represent the three activity areas (i.e. container alley, trailer alley and trailer parking). Please note that each figure has 12 calibration points. The calibration focused a greater weight on matching the two "4 Week" calibration points in each activity categories. These were the longest duration calibration points which provided the greatest amount of rainfall and associated washoff interaction. Washoff interaction is extremely important in the application of the calibrated model which was to simulate the stormwater pollutant loads that would occur annually from a hypothetical marine container storage yard.

ANNUAL TSS LOAD REDUCTIONS FROM BMP'S

Using the calibrated model parameters from each of various activity areas, average annual Total Suspended Solids (TSS) loadings were simulated using an average year of rainfall events.

Average Rainfall Year Development

Rather than simulate many runs using many years and summarize the extensive results, a long rainfall record was processed into many years of events, which were then evaluated. The twelve "best" months were combined to synthesize an average year that was used for the annual runs to evaluate different BMP practices.

High quality, continuous hourly rainfall data had been collected and recorded by NOAA at SeaTac International Airport since 1965. Data through 1993 was obtained on CD-ROM and used in this study. Hourly rainfall was processed using RAINEV into discrete events that were summarized by the following parameters for each month of each year:

1. Number of events
2. Total duration of events
3. Total depth of events
4. Maximum hourly rainfall
5. Average intensity (i.e. total depth/total duration)
6. Average dry time preceding events

These were then analyzed graphically in a spreadsheet month by month. Figure 8 shows an example for April. Each statistic for each year was compared to its average for all years. The absolute "error" or "departure from the mean" was graphed by year, with emphasis on the total monthly depth. Months that closely approximate the mean were found by looking for years where all data points (i.e. "errors") neared "0". In this example, statistics for April of 1975 clearly are nearest those for all Aprils, since all lines are very close to "0".

In this manner, each of the twelve months was examined and the "best" month for each month was found. The hourly data for each were then combined to create a representative, average year, which was analyzed by RAINEV to generate the events used by SIMPTM in its average annual runs. RAINEV is a rainfall analysis program included in the SIMPTM package.

Table 8 presents the resulting composition and compares the monthly, seasonal, and annual rainfall depths for the whole record to those for this average year and those resulting if just an actual year like 1980 (rather than individual months) had been used. The average year is clearly more representative.

Figure 9 supports this conclusion by graphing the monthly rainfall depths associated with each of these three potential data sets. The average year monthly depths are nearly indistinguishable from the averages for the entire record.

TSS Loadings

Using the calibrated model parameters from each of three activity areas, the model was used to simulate the sediment and associated pollutant accumulation and washoff that will occur over an average year from a hypothetical 10-acre container yard. As presented above, the average precipitation year was obtained from a detailed statistical analysis of a 29 year precipitation record at SeaTac International Airport.

The model simulations dealt with alternative frequencies of sweeping (e.g. daily to monthly) and catchbasin cleaning (monthly to annually), along with alternative sizes of catchbasins (e.g. normal, at 10 cubic feet per 15,000 ft^2 of area or enlarged, at 37.8 cubic feet for the same area). The results of these simulations are presented in Tables 9, 10, 11, and 12.

As shown on these tables, the hypothetical 10-acre composite site or container storage yard was composed of five different activity areas. Area "A" was 5.0 acres in size and consisted of trailer and container alleys, roadways, parking lots, loading docks and entrance and exit areas. All of this area was assumed to be accessible to sweeping or "sweepable" and all of this area was assumed to be accessible to rainfall or "washable". Area "C" was 0.5 acres in size and consisted of container storage areas that are both "sweepable" and "washable" which was only half of the actual container storage area (i.e. 1.0 acre) within the composite site.

Area "T" was trailer parking and storage areas which were only 8% "sweepable" and 21.8% "washable". This 21.8% represents the 8% that is assumed vacant at any given time and the remaining 13.8% which is the area between the parked trailers. Area "X" is the "unsweepable" fringe area (i.e. 0.5 areas) between the ends of the trailer rows which is assumed to be "washable". Area "O" represents those areas (i.e. 3.31 acres total) that the model assumes runoff with no pollutant contributions from which includes roof tops (i.e. 1.0 acre), the tops of stored containers (i.e. 0.5 acres), and the tops of parked trailers (i.e. 1.81 acres).

The results clearly showed that sweeping could provide some significant reductions in TSS loadings especially at scheduled frequencies of seven days or less. Also, the simulated reductions suggested enlarged catchbasins or large inlet sumps with 37.8 ft^3/15,000 sf do not appear to be worth the added cost. Normal catchbasins with 10.0 ft^3/15,000 sf cleaned annually appear to provide reasonable good TSS load reductions when combined with frequent high efficiency sweeping.

Dry Versus Damp Pavement Sweeping

As stated earlier, the computer simulation considered the effect of sweeping frequency on the reduction of pollutants in individual storms that occur throughout an average precipitation year. It also considered the effects of rainfall periods on both the ability to sweep and the efficiency of the sweeper. Rainfall periods are of three types — dry period (high sweeping efficiency), rainy periods that produce runoff (sweeping cannot occur), and rain periods that cause damp pavement conditions without runoff (sweeping occurs but at reduced pickup efficiency).

Unfortunately, the computer model does not have the ability to alter the basic performance characteristics of the sweeper during a simulation. So, two sets of model simulations were obtained —

one assumed the performance characteristics of damp pavement sweeping and the other assumed those of dry sweeping.

For the damp pavement or dry pavement conditions noted in Tables 9 - 12, this means that when sweeping was scheduled to occur and provided that runoff is not occurring at that time (i.e. sweeping would not be simulated if runoff conditions exist), SIMPTM assumes the pickup performance characteristics of damp pavement or dry pavement, respectively, throughout the entire annual simulation. Since damp pavement conditions will occur on occasion, throughout the year along with dry pavement conditions, one can probably assume that the actual pollutant removal effectiveness from the model simulation would be somewhere in between the two removals shown for each condition listed.

Dry pavement sweeping pickup performance was based on the SSmin and SSeff characteristics found for the Enviro Whirl I dry sweeper tests. Damp pavement sweeping pickup performance was based on the SSmin and SSeff characteristics for the tandem operation which included the use of a tanker truck that sprayed water on the street surface just before it was swept by both a mechanical and vacuum sweeper.

Please note that the modeled difference between these two sweeping conditions (i.e. damp or dry pavement) is not very significant. This is due primarily to the overland washoff hydraulic characteristics which are very sensitive to the sweepers removal of coarse particles (i.e. 250 microns or greater) which is quite good in both cases. Since both sweeping operations (i.e. damp versus dry) are also considered quite effective in the pickup of mid range particles (i.e. 63 to 250 microns), the net effects of the greater pickup performance of the fine material (i.e. less than 63 microns) is not that great.

Dramatic changes in performance may occur between damp and dry sweeping when occasional concentrated runoff conditions like gutter flow or swale flow are present on the site. It is important that the Port uses for any field testing or actual program implementation the most efficient pavement sweeper that is known to exist which is the Enviro Whirl dry sweeper manufactured by Enviro Whirl Technologies, Inc. out of Centralia, Illinois.

ANNUAL POLLUTANT LOADS AND CONCENTRATIONS

Sediment washoff was related to washoff of other pollutants by mass-fractions or potency factors assigned to each of the eight particle size groups of accumulated sediment. These are generally set from observed fractions of accumulated sediment or from observed sediment washed off during sampled events. Since no washoff data was obtained, the pollutant simulations were based on the chemical analyses of the collected samples.

As discussed earlier, chemical analyses of yard dirt samples were conducted for metals, COD, TKN, TP, Ortho Phosphorus and TPH. It was determined that in addition to TSS the SIMPTM pollutant simulations would include copper, lead, zinc and phosphorus. The simulation of the particulate or suspended fraction of each pollutant was based on the mean potency factors found in the yard dirt analyses. The dissolved fraction of each pollutant was based on the following assumptions: 50% of the copper, phosphorus and zinc washoff at any given time is dissolved whereas only 20% of the lead is dissolved. It was assumed that the sweeper could remove dissolved pollutants but the catchbasins and wet vaults could not remove any dissolved pollutants.

The results of these model simulations which will be presented in this section, were then compared to the results obtained from the simulated performance of a wet vault designed to the Puget Sound water quality criteria. This criteria would require a 50,000 ft^3 stormwater storage area needed to serve the

hypothetical 10 acre container yard. The sediment and associated pollutant removals of the wet vault were based on a set of modified Stoke's Law equations needed to compute settling velocities for various sizes of transported sediments. These calculations were processed in a spreadsheet using outputs from the SIMPTM model as the starting point. These model simulations assumed no sweeping but they did assume that enlarged catchbasins would be used and would be cleaned at the same frequency that the comparable sweeping with no wet vault runs had assumed. By holding size and cleaning frequency of the catchbasins the same, the results will truly show the anticipated pollutant reductions resulting from alternative levels of sweeping.

The results of the SIMPTM pollutant simulations for the average rainfall year are presented in Tables 13 through 16. As stated earlier, both dry pavement sweeping and damp pavement sweeping were simulated separately. Sweeping intervals were held to 30, 15, 7, 4 and 3 days, respectively.

The results of the study show that high efficiency sweeping will provide significant reductions in annual pollutant loadings of the five pollutants that were simulated which included total suspended solids (TSS), total copper (Cu), total phosphorus (T-P), total lead (T-Pb), and total zinc (T-Zn). These reductions range from an estimated 29% to 78% depending on the sweeping frequency, the catchbasin size and the specified pollutant.

When compared to the estimated pollutant removal of wet vaults (in combination with enlarged catchbasins), sweeping (in combination with normal or enlarged catchbasins) is projected to equal or exceed the removals obtained for total copper, total phosphorus and total zinc. However, wet vaults (in combination with enlarged catchbasins) are projected to provide a higher level of pollutant reduction for TSS and total lead at all simulated sweeping frequencies.

It should be noted that the SIMPTM simulation results presented in Tables 13 - 16 assumed that the sediment trapping catchbasins would be cleaned twice a year. This is an oversight since the difference between annual cleaning and twice annual cleaning is not very significant (i.e. 3 to 5%) as can be seen on the previously presented Tables 10 and 12. In addition, enlarged catchbasins were used in conjunction with the wet vault simulations. Once again this was an oversight by KAI. Both of these oversights could have been corrected but the resources needed to rerun the simulations were not currently available. The important point is that during actual program implementation normal sized catchbasins should be used with annual cleaning.

STUDY CONCLUSIONS

The following study conclusions can be offered:

1. For marine cargo handling and storage facilities, the pollutant removals associated with high efficiency sweeping at a weekly frequency in combination with normal catchbasin inlets, cleaned annually is essentially equivalent to those removals obtained by wet vaults.

2. Sweeping appears to be competitive with wet vaults even though a sweeper is unable to access all of the container yard. A sweeper is unable to sweep all of the site because of the presence of parked container trailers which cannot be moved to accommodate the needs of sweeping. Despite this limitation, sweepers may be essentially equivalent to wet vaults because of their high pickup efficiency, as well as their ability to remove dissolved pollutants.

3. Wet vaults are able to remove only sediment and the associated particulate pollutants. They are not able to remove dissolved pollutants. In contrast, high efficiency sweepers are able to remove dissolved pollutants since this fraction is contained in the accumulated sediment and can be removed by the sweeper during dry periods.

4. TSS is the characteristic of stormwater that is commonly used to compare the performance of alternative stormwater BMP's. Sweeping may be unable to achieve the high TSS removal efficiency simulated for wet vaults.

5. High efficiency sweeping may be more efficient than wet vaults in the removal of pollutants that tend to have a significant fraction that is dissolved such as copper, zinc, and phosphorus.

6. Wet vaults may be more efficient than high efficiency sweeping in the removal of pollutants that tend to have a significant fraction in the particulate state such as lead.

7. For high efficiency sweeping to be used as an alternative to wet vaults, the sweepable portions of the site must be swept at least weekly. For the Southwest Harbor Project mentioned earlier, this is anticipated to require the full-time dedication of at least one sweeper. The model simulations assumed that no sweeping will occur when runoff conditions exist. This is the actual practice that needs to be implemented in the field.

8. The efficiency of wet vaults in the study is calculated number based on modified Stoke's Law equation needed to compute settling velocities for various sizes of transported sediments . It is probable that wet vault efficiency is actually lower in practice. As a result, high efficiency sweeping may be considered an even more viable solution under actual field conditions.

Given the potential uncertainty of the model simulations presented in Tables 9 through 16, ranges of possible removal efficiencies are presented in Table 17. Also presented in Table 17 is the effect of sweeping frequency. With regard to sweeping frequency, the model suggests there may be significant benefit in sweeping weekly as opposed to biweekly. However, there may be little incremental benefit of sweeping twice weekly in comparison to weekly.

With regard to overall removal efficiencies, the model simulations presented in Tables 9 through 16 include two assumptions that are altered to provide the range in Table 17. First, the model assumes that the sweeper removes the same fractions of dissolved pollutants associated with particulate material during damp pavement conditions as it does during dry pavement conditions. This may not be the case since some of the dissolved fraction could be transported to the catchbasins before it can be swept. Secondly, the model assumes that material that accumulates under the trailers is not washed from the site since the topography of a container yard is such that stormwater typically does not flow beneath trailers except in extreme events. Regardless, material is moved from under the trailers into the alleyway. It is likely, however, that because of the overland flow condition, much of the material that is washed from beneath the trailers does not reach the catchbasins; rather, it settles on the pavement and could be removed by sweeping immediately after the event. Nonetheless, the low end of each range presented in Table 17 assumes that none of the dissolved pollutants are captured by sweeping during the damp pavement conditions sweeping and that the parked trailers do not block the potential transport of material beneath trailers so it is assumed to wash directly to the catchbasins. Hence, the vault is given the opportunity to capture this material whereas the sweeping is not.

With regard to wet vaults, SIMPTM predicts a removal efficiency of about 90% for TSS (Tables 13 through 16). This is greater than other predictive methods such as USEPA (1986). The USEPA methodology estimates removal efficiency using rainfall statistics and settling velocity data generated in the NURP study. Based on the concept of the volume specified by the Department of Ecology (1992), the estimated performance according the USEPA (1986) is about 75%. The ranges in the removal efficiencies for the wet vault in lower end of the ranges shown in Table 17 reflect somewhat more conservative assumptions as described in the report. What Table 17 still indicates is that ranges for sweeping are similar to the range for wet vaults for those pollutants that tend to be more dissolved.

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TABLE - 1

MECHANICAL SIEVE ANALYSIS OF SAMPLED SEDIMENT ACCUMULATIONS

Sample #	Date	Type	Area (sq.ft)	Total Wgt (gms)	Total Wgt (lbs/acre)	Particle Fraction Size Range (microns) Group 1 Size < 63	2 63-125	3 125-250	4 250-600	5 600-1000	6 1000-2000	7 2000-6370	8 > 6370
TP - Trailer Parking													
1	05/31/96	Initial	6000	15202.8	243.1	7.7%	22.9%	21.8%	16.6%	16.5%	6.7%	4.6%	3.1%
4	06/05/96	Week-1	2000	2059.5	98.8	10.0%	12.1%	12.7%	21.8%	14.3%	13.2%	11.2%	4.6%
11	06/14/96	Week-1	2000	2120.3	101.7	7.2%	14.7%	15.4%	21.8%	13.0%	11.3%	10.3%	6.4%
15	06/21/96	Week-1	2000	1115.5	53.5	12.3%	13.3%	13.2%	19.9%	14.1%	12.9%	10.5%	3.8%
18	06/27/96	Week-1	2000	1503	72.1	9.0%	10.4%	12.7%	21.5%	16.2%	17.0%	10.3%	3.0%
26	07/05/96	Week-1	2000	639.1	30.7	11.1%	13.2%	14.7%	21.8%	14.1%	12.9%	8.9%	3.3%
29	07/10/96	Week-1	2000	590	28.3	24.3%	21.6%	16.8%	17.8%	6.1%	6.0%	4.3%	1.1%
12	06/14/96	Week-2	2000	2919.6	140.1	8.5%	12.1%	11.0%	18.5%	15.6%	17.5%	12.0%	4.9%
20	06/27/96	Week-2	2000	1943.7	93.2	9.7%	16.5%	14.8%	23.4%	16.0%	13.3%	4.7%	1.4%
32	07/10/96	Week-2	2000	1431.4	68.7	19.4%	27.2%	15.1%	15.1%	8.9%	8.3%	4.0%	2.0%
36	07/30/96	Week-2	2000	1245.3	59.7	28.7%	20.2%	16.8%	17.8%	7.1%	5.8%	3.3%	0.3%
22	06/27/96	Week-4	2000	2658.5	127.5	5.4%	8.4%	11.7%	22.6%	19.2%	18.3%	11.2%	3.3%
38	07/30/96	Week-4	2000	2312.1	110.9	12.6%	15.1%	12.7%	18.6%	14.7%	15.4%	7.9%	2.9%
				Average	94.5	12.8%	16.0%	14.6%	19.8%	13.7%	12.2%	7.9%	3.1%
TA - Trailer Alley													
2	05/31/96	Initial	6000	3437.4	55.0	1.7%	3.4%	5.0%	12.2%	16.0%	25.9%	25.9%	9.9%
5	06/05/96	Week-1	2000	945.2	45.3	4.2%	7.3%	11.3%	21.5%	18.0%	20.2%	12.9%	4.5%
9	06/13/96	Week-1	2000	1140	54.7	1.1%	3.2%	9.2%	30.2%	23.7%	20.4%	10.1%	2.0%
14	06/21/96	Week-1	2000	359	17.2	0.7%	2.1%	8.6%	39.4%	23.8%	14.2%	5.4%	5.7%
17	06/27/96	Week-1	2000	529.9	25.4	0.3%	1.3%	7.4%	35.1%	22.9%	18.2%	10.2%	4.6%
25	07/05/96	Week-1	2000	538.1	25.8	0.4%	2.5%	11.9%	40.9%	20.2%	13.6%	8.0%	2.5%
28	07/10/96	Week-1	2000	389.9	18.7	1.4%	5.1%	17.1%	41.8%	14.8%	8.4%	4.9%	6.4%
10	06/13/96	Week-2	2000	1034.9	49.6	0.2%	1.0%	4.8%	24.4%	24.4%	26.1%	13.9%	5.1%
19	06/27/96	Week-2	2000	1066.4	51.2	0.3%	1.7%	7.2%	32.0%	25.2%	19.6%	10.4%	3.6%
31	07/10/96	Week-2	2000	465.6	22.3	1.3%	2.5%	7.3%	32.8%	21.2%	14.4%	9.6%	10.9%
37	07/30/96	Week-2	2000	3369.6	161.7	1.0%	3.1%	12.3%	36.7%	18.7%	12.5%	9.8%	5.9%
23	06/27/96	Week-4	2000	1955.5	93.8	0.8%	2.9%	10.0%	31.8%	23.4%	18.8%	8.5%	3.8%
39	07/30/96	Week-4	2000	3831	183.8	8.5%	15.2%	15.5%	24.7%	12.6%	9.8%	7.3%	6.3%
				Average	61.9	1.7%	4.0%	9.8%	31.0%	20.4%	17.1%	10.5%	5.5%
CA - Container Alley													
3	05/31/96	Initial	6000	6356.2	101.6	5.1%	20.0%	38.3%	18.6%	13.9%	3.1%	0.9%	0.2%
6	06/05/96	Week-1	2000	812.2	39.0	0.7%	2.9%	9.1%	29.3%	19.3%	22.0%	12.6%	4.0%
7	06/13/96	Week-1	2000	851.4	40.8	0.3%	1.1%	3.5%	27.7%	30.5%	24.8%	8.4%	3.7%
13	06/21/96	Week-1	2000	979	47.0	0.2%	1.2%	5.8%	34.3%	28.1%	19.7%	7.3%	3.2%
16	06/27/96	Week-1	2000	699.2	33.5	0.6%	3.4%	10.1%	29.5%	21.8%	20.2%	11.8%	2.8%
27	07/05/96	Week-1	2000	558.5	26.8	0.2%	1.2%	5.9%	31.9%	25.1%	20.9%	10.7%	4.0%
30	07/10/96	Week-1	2000	1282.3	61.5	1.0%	6.8%	21.0%	38.3%	15.2%	9.3%	4.7%	3.7%
8	06/13/96	Week-2	2000	1091	52.3	0.2%	1.1%	4.4%	25.3%	27.2%	26.8%	10.1%	5.0%
21	06/27/96	Week-2	2000	1404.5	67.4	0.4%	3.0%	9.2%	28.0%	25.4%	22.4%	8.2%	3.5%
33	07/10/96	Week-2	2000	1081.8	51.9	0.8%	7.0%	20.6%	35.8%	16.2%	10.9%	4.8%	3.9%
34	07/28/96	Week-2	2000	1530.4	73.4	0.4%	2.7%	8.7%	28.0%	25.3%	19.9%	5.5%	9.4%
24	06/27/96	Week-4	2000	1897.5	91.0	0.4%	2.5%	7.8%	26.5%	26.8%	25.7%	9.2%	1.1%
35	07/28/96	Week-4	2000	2440.6	117.1	0.5%	2.2%	7.5%	27.0%	23.0%	23.8%	7.5%	8.6%
				Average	61.8	0.8%	4.2%	11.7%	29.2%	22.9%	19.2%	7.8%	4.1%

TABLE - 2

CHEMICAL ANALYSIS OF SAMPLED SEDIMENT ACCUMULATIONS

Sample #	Concentration mg / kg (ppm) Fine	Medium	Coarse
Copper (Cu)			
1	190	120	23
2	190	280	36
3	220	330	1200
4	210	120	93
5	190	90	35
6	360	140	350
8	320	240	47
10	250	79	1100
12	240	120	99
22	210	220	69
23	220	84	55
24	370	110	48
Average	248	161	263
Lead (Pb)			
1	520	270	76
2	350	150	43
3	590	280	27
4	520	320	100
5	260	120	23
6	500	220	67
8	570	170	38
10	260	110	32
12	480	260	100
22	480	370	120
23	310	170	440
24	460	200	56
Average	442	220	94
Zinc (Zn)			
1	1800	2000	710
2	1500	1100	150
3	3000	2500	390
4	1800	2000	2000
5	1400	1200	170
6	3000	3200	1100
8	3600	2300	1700
10	1500	980	240
12	1900	1600	1900
22	1800	2200	1100
23	1700	920	280
24	3000	2700	560
Average	2167	1892	858
Total Petroleum Hydrocarbon (TPH)			
1	13400	7880	8390
2	11800	8050	1840
3	10500	4240	1160
4	7970	6670	5710
4-D	--	--	5770
5	5810	3920	1600
6	9420	8600	2480
8	9250	3320	1450
10	7200	3740	2085
12	9960	10000	6840
22	16500	11500	8040
23	8390	9120	5500
24	3700	4940	2100
Average	9492	6832	3935

TABLE - 3

CHEMICAL ANALYSIS OF SAMPLED SEDIMENT ACCUMULATIONS

Sample #	Concentration mg / kg (ppm)		
	Fine	Medium	Coarse
Chemical Oxygen Demand (COD)			
1	120000	90000	38000
2	120000	37000	6600
3	280000	53000	35000
22	140000	76000	58000
22-D	--	93000	--
23	160000	80000	25000
24	170000	26000	18000
Average	165000	65000	30100
Ortho Phosphorus (OP)			
1	640	410	190
2	520	340	140
3	460	390	240
3-D	--	250	--
22	540	390	240
23	510	370	260
24	470	320	230
24-D	--	300	--
Average	500	337.1	222
Total Kjeldahl Nitrogen (TKN)			
1	646	756	420
2	997	394	315
3	262	332	61.2
22	1560	927	525
23	1330	350	192
24	1140	367	140
Average	989.2	521	275.5

TABLE - 4

DISSOLVED METALS OBTAINED FROM TCLP

Sample #	Concentration mg / L (ppm)		
	Fine	Medium	Coarse
Copper			
CAF	0.15	--	--
7	--	0.16	0.07
TPF	0.15	--	--
9	--	0.27	0.1
11	0.57	0.51	0.14
18	0.38	0.43	0.11
37	0.08	--	--
19	--	0.56	0.11
39	0.05	--	--
21	--	0.14	0.1
Average	0.23	0.35	0.11
Lead			
CAF	0.5	--	--
7	--	1.3	2.5
TPF	ND	--	--
9	--	0.4	1.1
11	ND	ND	0.6
18	ND	0.3	ND
37	ND	--	--
19	--	0.6	ND
39	ND	--	--
21	--	0.5	1.1
Average	0.5	0.6	1.3
Zinc			
CAF	61	--	--
7	--	59	10
TPF	12	--	--
9	--	23	3.3
11	26	38	17
18	27	36	36
37	ND	--	--
19	--	18	5.1
39	ND	--	--
21	--	55	ND
Average	31.5	38.2	14.3

Notes:
TCLP - Total Contaminant Leachate Potential
CAF - Container Area Fines
(Consists of: 7F, 13F, 16F, 21F, 27F, 30F, & 33F)
TPF - Trailer Parking Fines
(Consists of: 9F, 14F, 17F, 19F, 25F, 28F & 31F)
ND - Not Detected
Average of samples above the detection limits

TABLE - 5

CHEMICAL ANALYSIS OF SAMPLED SEDIMENT ACCUMULATIONS

Sample #	Concentration mg / kg (ppm)		
	Fine	Medium	Coarse
Total Phosphorus			
CAF	450	--	--
7	--	310	500
TPF	350	--	--
9	--	320	280
11	710	490	230
18	670	450	330
37	580	--	--
19	--	360	200
19-D	--	350	--
39	620	--	--
21	--	270	250
Average	563	364	298
Ortho Phosphorus			
CAF	460	--	--
7	--	310	280
TPF	590	--	--
9	--	370	220
11	680	490	230
18	680	480	340
37	610	--	--
19	--	490	200
39	640	--	--
21	--	340	270
Average	610	413	257

Notes: CAF - Container Area Fines
(Consists of: 7F, 13F, 16F, 21F, 27F, 30F & 33F)
TPF - Trailer Parking Fines
(Consists of: 9F, 14F, 17F, 19F, 25F, 28F & 31F)

Table 6 Calibrated SSmin Comparison

Particle Size Group	Size Range *microns*	*Street Sweeping Technology* NURP Mech.	Newer Mech.	Tandem Sweeping	Regenerative Air	Enviro-Whirl
1	<63	9.0	5.8	2.0	0.0	0.0
2	-125	12.0	5.8	2.0	0.0	0.0
3	-250	18.0	5.3	2.3	0.9	0.0
4	-600	18.0	2.5	2.3	1.9	0.0
5	-1000	12.0	0.4	0.8	0.7	0.0
6	-2000	4.2	0.5	0.6	0.7	0.0
7	-6370	3.6	0.3	0.5	0.0	0.0
8	6370	1.8	0.0	0.0	0.0	0.0

Data from various studies, minimum pounds per paved acre remaining after street sweeping

Table 7 Calibrated Marginal Efficiencies - SSeff

Particle Size Group	Size Range *microns*	*Street Sweeping Technology* NURP Mech.	Newer Mech.	Tandem Sweeping	Regenerative Air	Enviro-Whirl
1	<63	44%	100%	93%	32%	70%
2	-125	52%	100%	95%	71%	77%
3	-250	47%	92%	93%	94%	84%
4	-600	50%	57%	89%	100%	88%
5	-1000	55%	48%	84%	100%	90%
6	-2000	60%	59%	88%	100%	91%
7	-6370	78%	81%	98%	94%	92%
8	>6370	79%	70%	87%	92%	96%

Data from various studies, marginal removal rate *only* for accumulations *greater* than SSmin

TABLE 8

REPRESENTATIVE YEAR COMPOSITION

Sea-Tac International Airport NOAA Hourly Rain Gage

Period	Year Used in Model	Rainfall Depths (inches) Model	Average	1980 Only
Annual		**35.77**	**35.90**	**35.38**
January	1976	5.55	5.28	4.23
February	1974	4.00	3.95	4.68
March	1988	3.75	3.62	2.10
April	1975	2.49	2.46	3.23
May	1976	1.61	1.62	0.97
June	1974	1.25	1.30	1.77
July	1979	0.73	0.79	0.46
August	1972	1.13	1.12	0.64
September	1988	1.75	1.83	1.43
October	1989	2.98	3.07	1.54
November	1982	5.31	5.33	6.94
December	1987	5.22	5.52	7.39

Notes:

Analysis based on 29-year period from 1965 through 1993 (inclusive)
Rainfall event tabulation performed using RAINEV (r)
1980 was the overall best-fit for an entire annual period (the 'old' method)
Fit determined visually by comparing deviations from monthly means for:

- Number of events
- Total duration of events
- Total depth of events
- Average maximum rainfall intensity during each events
- Average intensity of events
- Average antecedent dry time

Table 9

ANNUAL ALTERNATIVE TSS UNIT LOADINGS - DRY SWEEPING

Seattle Pier 5, Lbs per acre (or paved acre) per year, varied sweeping or trapping

Sweeping Interval (days)	Envirowhirl Sweeping with Varied Use of Sediment Trapping Catch Basins												
	Not Used	Normal Inlet Sumps (10 cuft / 15,000 sft) Cleaning Interval (days)						Large Inlet Sumps (37.8 cuft / 15,000 sft) Cleaning Interval (days)					
		360	180	90	60	30	1	360	180	90	60	30	1
A - Pier 5 Trailer and Container Alleys, Roadways, and Parking Lots, lbs per acre													
0	2505	1655	1335	1324	1321	1319	1317	1293	1291	1290	1290	1290	1289
90	2307	1467	1228	1219	1217	1215	1213	1191	1189	1188	1188	1188	1188
60	2290	1450	1219	1211	1208	1207	1205	1182	1181	1180	1180	1179	1179
30	2024	1184	1076	1070	1068	1067	1066	1045	1044	1043	1043	1043	1043
15	1621	879	862	859	858	857	856	837	837	836	836	836	836
7	1076	576	572	571	570	570	569	556	556	556	556	556	556
4	977	521	518	517	517	517	516	505	504	504	504	504	504
3	596	318	317	316	316	316	316	308	308	308	308	308	308
2	469	251	250	250	250	250	250	243	243	243	243	243	243
1	251	135	134	134	134	134	134	131	130	130	130	130	130
C - Pier 5 Container Storage Areas, lbs per acre (50% Sweepable & Washable)													
0	3935	3074	2356	2270	2255	2243	2234	2148	2134	2129	2127	2125	2124
90	3813	2961	2279	2199	2185	2174	2164	2081	2067	2062	2061	2059	2058
60	3767	2915	2243	2174	2159	2149	2139	2056	2043	2038	2037	2035	2034
30	3573	2721	2117	2061	2049	2039	2030	1950	1938	1934	1932	1931	1930
15	3251	2399	1916	1877	1867	1859	1852	1775	1765	1762	1761	1760	1758
7	2563	1712	1498	1479	1474	1469	1464	1399	1393	1391	1390	1392	1389
4	2273	1427	1323	1311	1306	1302	1298	1239	1235	1233	1233	1232	1232
3	1798	1074	1045	1037	1035	1032	1030	980	978	977	977	976	976
2	1470	870	854	850	848	846	845	803	801	800	800	800	800
1	877	515	511	509	509	508	508	480	480	479	479	479	479
T - Pier 5 Trailer Parking & Storage Areas, lbs per acre (8% Sweepable, 21.8% Washable)													
0	4062	3202	2452	2349	2331	2318	2308	2220	2204	2199	2197	2195	2193
90	4026	3168	2428	2327	2310	2298	2287	2200	2185	2179	2177	2175	2174
60	4007	3148	2410	2317	2299	2287	2276	2190	2174	2169	2167	2165	2164
30	3933	3075	2359	2273	2257	2245	2235	2149	2134	2129	2127	2126	2124
15	3808	2950	2274	2201	2186	2175	2165	2081	2067	2062	2060	2059	2057
7	3481	2624	2061	2010	1998	1989	1981	1900	1890	1886	1884	1885	1882
4	3252	2397	1916	1877	1867	1859	1852	1775	1766	1762	1761	1760	1759
3	2982	2128	1750	1720	1712	1705	1699	1627	1619	1616	1615	1614	1613
2	2650	1798	1549	1529	1522	1517	1512	1445	1440	1437	1437	1436	1435
1	1957	1178	1137	1128	1125	1122	1119	1066	1064	1062	1062	1061	1061
X - Pier 5 Unsweepable Fringe between Ends of Trailers, lbs per acre													
0	3832	2971	2282	2207	2193	2182	2173	2090	2076	2071	2070	2068	2067
COMPOSITE SITE, total lbs - 10-acre site coposed of: (5.0 A, 0.5 X, 0.5 C, 0.69 T, 3.31 "0")													
0	19209	13509	10685	10477	10438	10408	10382	10115	10080	10067	10062	10059	10055
90	18137	12487	10096	9903	9867	9838	9814	9557	9524	9512	9508	9504	9501
60	18016	12366	10021	9841	9804	9776	9752	9495	9462	9450	9446	9443	9439
30	16536	10887	9207	9051	9019	8994	8972	8726	8697	8687	8683	8679	8676
15	14272	9118	7978	7854	7827	7805	7785	7555	7530	7521	7518	7515	7512
7	10978	7032	6171	6082	6062	6046	6032	5837	5818	5811	5809	5807	5805
4	10181	6458	5716	5640	5622	5608	5594	5412	5396	5390	5388	5386	5384
3	7853	5079	4455	4392	4377	4365	4354	4200	4186	4181	4179	4178	4176
2	6827	4415	3888	3833	3820	3810	3801	3660	3648	3644	3642	3641	3639
1	4960	3229	2854	2808	2799	2791	2784	2673	2664	2661	2660	2658	2657

Note.

Dry condition sweeping parameters use values observed for the Envirowhirl I sweeper.

Table 10

ANNUAL ALTERNATIVE TSS LOAD REDUCTIONS - DRY SWEEPING

Seattle Pier 5, Percentage reduced per year by increased sweeping or inlet trapping

Sweeping Interval (days)	Envirowhirl Sweeping with Varied Use of Sediment Trapping Catch Basins												
	Not Used	Normal Inlet Sumps (10 cuft / 15,000 sft) Cleaning Interval (days)						Large Inlet Sumps (37.8 cuft / 15,000 sft) Cleaning Interval (days)					
		360	180	90	60	30	1	360	180	90	60	30	1
A - Pier 5 Trailer and Container Alleys, Roadways, and Parking Lots													
0	*Reference*	34%	47%	47%	47%	47%	47%	48%	48%	48%	49%	49%	49%
91	8%	41%	51%	51%	51%	51%	52%	52%	53%	53%	53%	53%	53%
61	9%	42%	51%	52%	52%	52%	52%	53%	53%	53%	53%	53%	53%
30	19%	53%	57%	57%	57%	57%	57%	58%	58%	58%	58%	58%	58%
15	35%	65%	66%	66%	66%	66%	66%	67%	67%	67%	67%	67%	67%
7	57%	77%	77%	77%	77%	77%	77%	78%	78%	78%	78%	78%	78%
4	61%	79%	79%	79%	79%	79%	79%	80%	80%	80%	80%	80%	80%
3	76%	87%	87%	87%	87%	87%	87%	88%	88%	88%	88%	88%	88%
2	81%	90%	90%	90%	90%	90%	90%	90%	90%	90%	90%	90%	90%
1	90%	95%	95%	95%	95%	95%	95%	95%	95%	95%	95%	95%	95%
C - Pier 5 Container Storage Areas (50% Sweepable & Washable)													
0	*Reference*	22%	40%	42%	43%	43%	43%	45%	46%	46%	46%	46%	46%
91	3%	25%	42%	44%	44%	45%	45%	47%	47%	48%	48%	48%	48%
61	4%	26%	43%	45%	45%	45%	46%	48%	48%	48%	48%	48%	48%
30	5%	31%	46%	48%	48%	48%	48%	50%	51%	51%	51%	51%	51%
15	17%	39%	51%	52%	53%	53%	53%	55%	55%	55%	55%	55%	55%
7	35%	56%	62%	62%	63%	63%	63%	64%	65%	65%	65%	65%	65%
4	42%	64%	66%	67%	67%	67%	67%	69%	69%	69%	69%	69%	69%
3	54%	73%	73%	74%	74%	74%	74%	75%	75%	75%	75%	75%	75%
2	63%	78%	78%	78%	78%	78%	79%	80%	80%	80%	80%	80%	80%
1	78%	87%	87%	87%	87%	87%	87%	88%	88%	88%	88%	88%	88%
T - Pier 5 Trailer Parking & Storage Areas (8% Sweepable, 21.8% Washable)													
0	*Reference*	21%	40%	42%	43%	43%	43%	45%	46%	46%	46%	46%	46%
91	1%	22%	40%	43%	43%	43%	44%	46%	46%	46%	46%	46%	46%
61	1%	23%	41%	43%	43%	44%	44%	46%	46%	47%	47%	47%	47%
30	3%	24%	42%	44%	44%	45%	45%	47%	47%	48%	48%	48%	48%
15	5%	27%	44%	46%	46%	46%	47%	49%	49%	49%	49%	49%	49%
7	13%	35%	49%	51%	51%	51%	51%	53%	53%	54%	54%	54%	54%
4	20%	41%	53%	54%	54%	54%	54%	56%	57%	57%	57%	57%	57%
3	27%	48%	57%	58%	58%	58%	58%	60%	60%	60%	60%	60%	60%
2	35%	56%	62%	62%	63%	63%	63%	64%	65%	65%	65%	65%	65%
1	52%	71%	72%	72%	72%	72%	72%	74%	74%	74%	74%	74%	74%
X - Pier 5 Unsweepable Fringe between Ends of Trailers													
0	*Reference*	22%	40%	42%	43%	43%	43%	45%	46%	46%	46%	46%	46%
COMPOSITE SITE - 10-acre site coposed of: (5.0 A, 0.5 X, 0.5 C, 0.69 T, 3.31 "0")													
0	*Reference*	30%	44%	45%	46%	46%	46%	47%	48%	48%	48%	48%	48%
91	5%	35%	47%	48%	48%	49%	49%	50%	50%	50%	50%	50%	50%
61	5%	35%	47%	48%	49%	49%	49%	50%	50%	50%	50%	50%	50%
30	12%	42%	51%	52%	52%	52%	52%	54%	54%	54%	54%	54%	54%
15	23%	51%	57%	58%	58%	58%	58%	59%	59%	59%	60%	60%	60%
7	38%	60%	65%	66%	66%	66%	66%	67%	67%	67%	67%	67%	67%
4	42%	63%	67%	68%	68%	68%	68%	69%	69%	69%	69%	69%	69%
3	53%	70%	73%	73%	73%	74%	74%	75%	75%	75%	75%	75%	75%
2	57%	73%	76%	76%	76%	76%	76%	77%	77%	77%	77%	77%	77%
1	66%	78%	80%	80%	80%	81%	81%	81%	81%	81%	81%	81%	81%

Note:

Dry condition sweeping parameters use values observed for the Envirowhirl I sweeper.

Table 11

ANNUAL ALTERNATIVE TSS LOADINGS - DAMP SWEEPING

Seattle Pier 5, Lbs per year, varied sweeping or trapping

Sweeping Interval (days)	Tandem (Damp) Sweeping with Varied Use of Sediment Trapping Catch Basins												
	Not Used	Normal Inlet Sumps (10 cuft / 15,000 sft) Cleaning Interval (days)						Large Inlet Sumps (37.8 cuft / 15,000 sft) Cleaning Interval (days)					
		360	180	90	60	30	1	360	180	90	60	30	1
A - Pier 5 Trailer and Container Alleys, Roadways, and Parking Lots, lbs per acre													
0	2505	1655	1335	1324	1321	1319	1317	1293	1291	1290	1290	1290	1289
90	2325	1483	1238	1229	1227	1225	1224	1200	1199	1198	1198	1197	1197
60	2319	1477	1236	1227	1225	1223	1221	1198	1196	1195	1195	1195	1195
30	2082	1241	1110	1103	1102	1100	1099	1077	1075	1075	1074	1074	1074
15	1714	939	916	912	911	910	909	888	887	887	887	887	886
7	1255	683	676	674	673	673	672	654	654	653	653	653	653
4	1160	632	626	624	624	623	623	606	605	605	605	605	605
3	824	453	451	450	449	449	449	434	434	434	434	434	433
2	720	399	397	396	396	396	395	381	381	381	381	381	381
1	537	304	302	302	302	302	302	288	288	288	288	288	288
C - Pier 5 Container Storage Areas, lbs per acre (50% Sweepable & Washable)													
0	3935	3074	2356	2270	2255	2243	2234	2148	2134	2129	2127	2125	2124
90	3815	2962	2280	2200	2185	2174	2165	2082	2068	2063	2062	2060	2059
60	3774	2921	2246	2177	2162	2151	2142	2059	2046	2041	2040	2038	2037
30	3586	2734	2123	2066	2054	2044	2036	1955	1944	1940	1938	1937	1936
15	3273	2420	1926	1886	1877	1868	1861	1785	1775	1772	1771	1769	1768
7	2629	1777	1532	1512	1506	1501	1497	1432	1426	1424	1423	1422	1422
4	2362	1515	1372	1358	1353	1349	1345	1285	1281	1279	1279	1278	1278
3	1922	1152	1114	1106	1103	1100	1097	1046	1043	1042	1042	1042	1041
2	1631	967	946	940	938	936	934	889	887	886	886	886	885
1	1110	654	647	645	644	643	642	608	607	607	606	606	606
T - Pier 5 Trailer Parking & Storage Areas, lbs per acre (8% Sweepable, 21.8% Washable)													
0	4062	3202	2452	2349	2331	2318	2308	2220	2204	2199	2197	2195	2193
90	4027	3168	2429	2328	2311	2298	2287	2201	2185	2179	2178	2176	2174
60	4009	3150	2411	2317	2300	2288	2277	2191	2175	2170	2168	2166	2165
30	3937	3079	2361	2275	2259	2247	2237	2151	2136	2131	2129	2128	2126
15	3816	2958	2278	2204	2190	2178	2169	2084	2071	2066	2064	2063	2061
7	3504	2647	2073	2021	2009	2000	1991	1912	1901	1897	1895	1894	1893
4	3287	2432	1935	1895	1884	1876	1869	1792	1783	1779	1778	1777	1776
3	3032	2177	1776	1745	1737	1730	1724	1652	1644	1641	1640	1639	1638
2	2723	1870	1589	1567	1560	1555	1550	1483	1477	1475	1474	1473	1472
1	2086	1268	1211	1200	1196	1193	1190	1135	1132	1131	1130	1130	1129
X - Pier 5 Unsweepable Fringe between Ends of Trailers, lbs per acre													
0	3832	2971	2282	2207	2193	2182	2173	2090	2076	2071	2070	2068	2067
COMPOSITE SITE, total lbs - 10-acre site coposed of: (5.0 A, 0.5 X, 0.5 C, 0.69 T, 3.31 "0")													
0	19209	13509	10685	10477	10438	10408	10382	10115	10080	10067	10062	10059	10055
90	18226	12568	10148	9954	9917	9889	9864	9605	9573	9560	9556	9552	9549
60	18163	12505	10107	9925	9887	9860	9835	9575	9542	9530	9526	9522	9518
30	16837	11180	9382	9223	9190	9164	9141	8889	8860	8848	8845	8842	8838
15	14755	9434	8257	8128	8100	8077	8057	7816	7791	7781	7778	7775	7772
7	11921	7618	6719	6625	6603	6585	6570	6350	6330	6323	6320	6318	6315
4	11167	7078	6293	6212	6192	6176	6161	5952	5934	5928	5925	5923	5921
3	9090	5829	5177	5109	5093	5080	5069	4877	4863	4857	4855	4853	4851
2	8210	5254	4695	4635	4622	4610	4600	4418	4405	4400	4398	4396	4395
1	6596	4206	3812	3764	3753	3744	3736	3573	3563	3559	3558	3556	3555

Note:

Damp condition sweeping parameters use values observed for tandem sweeping (standard flush / mechanical followed by vaccuum unit).

Table 12

ANNUAL ALTERNATIVE TSS LOAD REDUCTIONS - DAMP SWEEPING

Seattle Pier 5, Percentage reduced per year by increased sweeping or inlet trapping

Sweeping Interval (days)	Tandem (Damp) Sweeping with Varied Use of Sediment Trapping Catch Basins												
	Not Used	Normal Inlet Sumps (10 cuft / 15,000 sft) Cleaning Interval (days)						Large Inlet Sumps (37.8 cuft / 15,000 sft) Cleaning Interval (days)					
		360	180	90	60	30	1	360	180	90	60	30	1
A - Pier 5 Trailer and Container Alleys, Roadways, and Parking Lots													
0	*Reference*	34%	47%	47%	47%	47%	47%	48%	48%	48%	49%	49%	49%
91	7%	41%	51%	51%	51%	51%	51%	52%	52%	52%	52%	52%	52%
61	7%	41%	51%	51%	51%	51%	51%	52%	52%	52%	52%	52%	52%
30	17%	50%	56%	56%	56%	56%	56%	57%	57%	57%	57%	57%	57%
15	32%	62%	63%	64%	64%	64%	64%	65%	65%	65%	65%	65%	65%
7	50%	73%	73%	73%	73%	73%	73%	74%	74%	74%	74%	74%	74%
4	54%	75%	75%	75%	75%	75%	75%	76%	76%	76%	76%	76%	76%
3	67%	82%	82%	82%	82%	82%	82%	83%	83%	83%	83%	83%	83%
2	71%	84%	84%	84%	84%	84%	84%	85%	85%	85%	85%	85%	85%
1	79%	88%	88%	88%	88%	88%	88%	88%	88%	89%	89%	89%	89%
C - Pier 5 Container Storage Areas (50% Sweepable & Washable)													
0	*Reference*	22%	40%	42%	43%	43%	43%	45%	46%	46%	46%	46%	46%
91	3%	25%	42%	44%	44%	45%	45%	47%	47%	48%	48%	48%	48%
61	4%	26%	43%	45%	45%	45%	46%	48%	48%	48%	48%	48%	48%
30	9%	31%	46%	47%	48%	48%	48%	50%	51%	51%	51%	51%	51%
15	17%	38%	51%	52%	52%	53%	53%	55%	55%	55%	55%	55%	55%
7	33%	55%	61%	62%	62%	62%	62%	64%	64%	64%	64%	64%	64%
4	40%	62%	65%	65%	66%	66%	66%	67%	67%	67%	68%	68%	68%
3	51%	71%	72%	72%	72%	72%	72%	73%	73%	74%	74%	74%	74%
2	59%	75%	76%	76%	76%	76%	76%	77%	77%	77%	77%	77%	77%
1	72%	83%	84%	84%	84%	84%	84%	85%	85%	85%	85%	85%	85%
T - Pier 5 Trailer Parking & Storage Areas (8% Sweepable, 21.8% Washable)													
0	*Reference*	21%	40%	42%	43%	43%	43%	45%	46%	46%	46%	46%	46%
91	1%	22%	40%	43%	43%	43%	44%	46%	46%	46%	46%	46%	46%
61	1%	22%	41%	43%	43%	44%	44%	46%	46%	47%	47%	47%	47%
30	5%	24%	42%	44%	44%	45%	45%	47%	47%	48%	48%	48%	48%
15	9%	27%	44%	46%	46%	46%	47%	49%	49%	49%	49%	49%	49%
7	14%	35%	49%	50%	51%	51%	51%	53%	53%	53%	53%	53%	53%
4	19%	40%	52%	53%	54%	54%	54%	56%	56%	56%	56%	56%	56%
3	25%	46%	56%	57%	57%	57%	58%	59%	60%	60%	60%	60%	60%
2	33%	54%	61%	61%	62%	62%	62%	63%	64%	64%	64%	64%	64%
1	49%	69%	70%	70%	71%	71%	71%	72%	72%	72%	72%	72%	72%
X - Pier 5 Unsweepable Fringe between Ends of Trailers													
0	*Reference*	22%	40%	42%	43%	43%	43%	45%	46%	46%	46%	46%	46%
COMPOSITE SITE - 10-acre site coposed of: (5.0 A, 0.5 X, 0.5 C, 0.69 T, 3.31 "0")													
0	*Reference*	30%	44%	45%	46%	46%	46%	47%	48%	48%	48%	48%	48%
91	5%	35%	47%	48%	48%	49%	49%	50%	50%	50%	50%	50%	50%
61	5%	35%	47%	48%	49%	49%	49%	50%	50%	50%	50%	50%	50%
30	12%	42%	51%	52%	52%	52%	52%	54%	54%	54%	54%	54%	54%
15	23%	51%	57%	58%	58%	58%	58%	59%	59%	59%	60%	60%	60%
7	38%	60%	65%	66%	66%	66%	66%	67%	67%	67%	67%	67%	67%
4	42%	63%	67%	68%	68%	68%	68%	69%	69%	69%	69%	69%	69%
3	53%	70%	73%	73%	73%	74%	74%	75%	75%	75%	75%	75%	75%
2	57%	73%	76%	76%	76%	76%	76%	77%	77%	77%	77%	77%	77%
1	66%	78%	80%	80%	80%	81%	81%	81%	81%	81%	81%	81%	81%

Note:

Damp condition sweeping parameters use values observed for tandem sweeping (standard flush / mechanical followed by vaccuum unit).

Table 13

ANNUAL ALTERNATIVE POLLUTANT LOADS - DRY SWEEPING

Seattle Pier 5, Lbs per year

Pollutant Washoff		Baseline	Normal WQ Inlets					Enlarged WQ Inlets					WQ Vault
Average Annual Amounts		*No Trapping No Sweeping*	*Sweeping Interval (days)* 30	15	7	4	3	*Sweeping Interval (days)* 30	15	7	4	3	*With Enlarged Inlets No Sweeping*
Annual Dissolved Loads (Lbs)													
D-Cu	Copper	4.67	4.00	3.44	2.63	2.44	1.87	4.00	3.44	2.63	2.44	1.87	4.67
D-P	Phosphorus	13.49	11.61	10.01	7.70	7.14	5.50	11.61	10.01	7.70	7.14	5.50	13.49
D-Pb	Lead	0.29	0.25	0.22	0.18	0.17	0.13	0.25	0.22	0.18	0.17	0.13	0.29
D-Zn	Zinc	21.55	18.77	16.40	12.89	11.97	9.48	18.77	16.40	12.89	11.97	9.48	21.55
Annual Suspended Loads (Lbs)													
TSS		19209	8697	7530	5818	5396	4186	9207	7978	6171	5716	4455	1693
S-Cu	Copper	4.67	2.08	1.80	1.38	1.28	0.99	2.19	1.89	1.45	1.35	1.04	0.41
S-P	Phosphorus	13.49	6.09	5.27	4.07	3.78	2.93	6.44	5.58	4.32	4.00	3.11	1.19
S-Pb	Lead	2.61	1.27	1.12	0.90	0.83	0.67	1.40	1.24	0.99	0.92	0.75	0.26
S-Zn	Zinc	21.55	10.22	8.97	7.09	6.59	5.25	11.14	9.78	7.74	7.18	5.75	2.02
Annual TOTAL Loads (Lbs)													
TSS		19209	8697	7530	5818	5396	4186	9207	7978	6171	5716	4455	1693
T-Cu	Copper	9.33	6.09	5.24	4.02	3.72	2.86	6.19	5.33	4.09	3.79	2.91	5.07
T-P	Phosphorus	26.97	17.70	15.29	11.77	10.92	8.43	18.05	15.59	12.01	11.14	8.62	14.68
T-Pb	Lead	2.90	1.52	1.34	1.07	1.00	0.81	1.65	1.46	1.17	1.09	0.88	0.55
T-Zn	Zinc	43.10	29.00	25.38	19.98	18.56	14.73	29.91	26.19	20.63	19.15	15.23	23.57
Annual Percent Removals													
TSS		*reference*	55%	61%	70%	72%	78%	52%	58%	68%	70%	77%	91%
T-Cu	Copper	*reference*	35%	44%	57%	60%	69%	34%	43%	56%	59%	69%	46%
T-P	Phosphorus	*reference*	34%	43%	56%	60%	69%	33%	42%	55%	59%	68%	46%
T-Pb	Lead	*reference*	48%	54%	63%	65%	72%	43%	50%	60%	62%	70%	81%
T-Zn	Zinc	*reference*	33%	41%	54%	57%	66%	31%	39%	52%	56%	65%	45%

Notes:

WQ Inlet Parameters

Cleaning Intervals	Every 180 days *(Twice per year)*
Inlet Density	2.9 per acres *(one per 15,000 paved sq. ft.)*
Sump Storage Depth	3 ft.
Normal Sump Area	3.33 sq. ft.
Normal Sump Volume	10.0 cu. ft.
Enlarged Sump Area	12.6 sq. ft.
Enlarged Sump Volume	37.8 cu. ft.

Water Quality Vault Parameters

Vault Contributing Area	10 Acres (paved)
Vault Area	10,000 sq. ft.
Vault Depth	5 ft
Vault Volume	50,000 cu. ft.

Note - vault performance included the 100% runoff from the Roofs area.

Study Area Composition

Acres	Description
10.00	Total Study Area
5.00	A - Alleys - Pier 5 Trailer and Container Alleys, Roadways, and Parking Lots
0.50	C - Containers - Pier 5 Container Storage Areas (50% Sweepable & Washable)
0.69	T - Trailers - Pier 5 Trailer Parking & Storage Areas (8% Sweepable, 21.8% Washable)
0.50	X - Pier 5 Unsweepable Fringe between Ends of Trailers
3.31	Roofs - Contributed 100% Runoff but no pollutant loads

Dry condition sweeping parameters use values observed for the Envirowhirl I sweeper.

Table 14

ANNUAL ALTERNATIVE POLLUTANT CONCENTRATIONS - DRY SWEEPING

Seattle Pier 5, Lbs per year

Pollutant Washoff		Baseline	Normal WQ Inlets					Enlarged WQ Inlets					WQ Vault
Average Annual Values (Volume-weighted Event Mean Conc'c)		*No Trapping No Sweeping*	*Sweeping Interval (days)* 30	15	7	4	3	*Sweeping Interval (days)* 30	15	7	4	3	*With Enlarged Inlets No Sweeping*
Annual Dissolved Average (mg/l)													
D-Cu	Copper	0.032	0.027	0.023	0.018	0.017	0.013	0.027	0.023	0.018	0.017	0.013	0.032
D-P	Phosphorus	0.092	0.079	0.068	0.052	0.049	0.037	0.079	0.068	0.052	0.049	0.037	0.092
D-Pb	Lead	0.002	0.002	0.002	0.001	0.001	0.001	0.002	0.002	0.001	0.001	0.001	0.002
D-Zn	Zinc	0.147	0.128	0.112	0.088	0.082	0.065	0.128	0.112	0.088	0.082	0.065	0.147
Annual Suspended Average (mg/l)													
TSS		130.7	59.2	51.3	39.6	36.7	28.5	62.7	54.3	42.0	38.9	30.3	11.5
S-Cu	Copper	0.032	0.014	0.012	0.009	0.009	0.007	0.015	0.013	0.010	0.009	0.007	0.003
S-P	Phosphorus	0.092	0.041	0.036	0.028	0.026	0.020	0.044	0.038	0.029	0.027	0.021	0.008
S-Pb	Lead	0.018	0.009	0.008	0.006	0.006	0.005	0.010	0.008	0.007	0.006	0.005	0.002
S-Zn	Zinc	0.147	0.070	0.061	0.048	0.045	0.036	0.076	0.067	0.053	0.049	0.039	0.014
Annual TOTAL Average (mg/l)													
TSS		130.7	59.2	51.3	39.6	36.7	28.5	62.7	54.3	42.0	38.9	30.3	11.5
T-Cu	Copper	0.064	0.041	0.036	0.027	0.025	0.019	0.042	0.036	0.028	0.026	0.020	0.035
T-P	Phosphorus	0.184	0.120	0.104	0.080	0.074	0.057	0.123	0.106	0.082	0.076	0.059	0.100
T-Pb	Lead	0.020	0.010	0.009	0.007	0.007	0.005	0.011	0.010	0.008	0.007	0.006	0.004
T-Zn	Zinc	0.293	0.197	0.173	0.136	0.126	0.100	0.204	0.178	0.140	0.130	0.104	0.160
Annual Percent Removals													
TSS		*reference*	55%	61%	70%	72%	78%	52%	58%	68%	70%	77%	91%
T-Cu	Copper	*reference*	35%	44%	57%	60%	69%	34%	43%	56%	59%	69%	46%
T-P	Phosphorus	*reference*	34%	43%	56%	60%	69%	33%	42%	55%	59%	68%	46%
T-Pb	Lead	*reference*	48%	54%	63%	65%	72%	43%	50%	60%	62%	70%	81%
T-Zn	Zinc	*reference*	33%	41%	54%	57%	66%	31%	39%	52%	56%	65%	45%

Notes:

WQ Inlet Parameters

Cleaning Intervals	Every 180 days *(Twice per year)*
Inlet Density	2.9 per acres *(one per 15,000 paved sq. ft.)*
Sump Storage Depth	3 ft.
Normal Sump Area	3.33 sq. ft.
Normal Sump Volume	10.0 cu. ft.
Enlarged Sump Area	12.6 sq. ft.
Enlarged Sump Volume	37.8 cu. ft.

Water Quality Vault Parameters

Vault Contributing Area	10 Acres (paved)
Vault Area	10,000 sq. ft.
Vault Depth	5 ft
Vault Volume	50,000 cu. ft.

Note - vault performance included the 100% runoff from the Roofs area.

Study Area Composition

Acres	Description
10.00	Total Study Area
5.00	A - Alleys - Pier 5 Trailer and Container Alleys, Roadways, and Parking Lots
0.50	C - Containers - Pier 5 Container Storage Areas (50% Sweepable & Washable)
0.69	T - Trailers - Pier 5 Trailer Parking & Storage Areas (8% Sweepable, 21.8% Washable)
0.50	X - Pier 5 Unsweepable Fringe between Ends of Trailers
3.31	Roofs - Contributed 100% Runoff but no pollutant loads

Dry condition sweeping parameters use values observed for the Envirowhirl I sweeper.

Total Runoff volume is 54.0 Acre-feet (mg/l = 0.3677 * lbs/acreft)

Table 15

ANNUAL ALTERNATIVE POLLUTANT LOADS - DAMP SWEEPING

Seattle Pier 5, Lbs per year

Pollutant Washoff		Baseline	Normal WQ Inlets					Enlarged WQ Inlets					WQ Vault
Average Annual Amounts		No Trapping No Sweeping	Sweeping Interval (days) 30	15	7	4	3	Sweeping Interval (days) 30	15	7	4	3	With Enlarged Inlets No Sweeping
Annual Dissolved Loads (Lbs)													
D-Cu	Copper	4.67	4.07	3.56	2.85	2.66	2.15	4.07	3.56	2.85	2.66	2.15	4.67
D-P	Phosphorus	13.49	11.81	10.35	8.35	7.82	6.35	11.81	10.35	8.35	7.82	6.35	13.49
D-Pb	Lead	0.29	0.26	0.23	0.20	0.18	0.16	0.26	0.23	0.20	0.18	0.16	0.29
D-Zn	Zinc	21.55	19.16	17.03	14.16	13.33	11.19	19.16	17.03	14.16	13.33	11.19	21.55
Annual Suspended Loads (Lbs)													
TSS		19209	8860	7790	6330	5934	4863	9382	8257	6719	6293	5177	1693
S-Cu	Copper	4.67	2.12	1.86	1.50	1.40	1.13	2.23	1.95	1.57	1.47	1.20	0.41
S-P	Phosphorus	13.49	6.21	5.45	4.42	4.15	3.39	6.56	5.77	4.69	4.39	3.61	1.19
S-Pb	Lead	2.61	1.29	1.17	0.99	0.93	0.80	1.43	1.29	1.09	1.03	0.89	0.26
S-Zn	Zinc	21.55	10.44	9.32	7.81	7.35	6.22	11.38	10.17	8.52	8.02	6.81	2.02
Annual TOTAL Loads (Lbs)													
TSS		19209	8860	7790	6330	5934	4863	9382	8257	6719	6293	5177	1693
T-Cu	Copper	9.33	6.20	5.41	4.34	4.06	3.28	6.30	5.51	4.42	4.13	3.34	5.07
T-P	Phosphorus	26.97	18.02	15.80	12.77	11.96	9.74	18.38	16.12	13.04	12.21	9.96	14.68
T-Pb	Lead	2.90	1.55	1.40	1.18	1.12	0.96	1.69	1.52	1.29	1.22	1.05	0.55
T-Zn	Zinc	43.10	29.60	26.36	21.97	20.69	17.41	30.54	27.20	22.68	21.35	18.00	23.57
Annual Percent Removals													
TSS		*reference*	*54%*	59%	67%	69%	75%	*51%*	57%	65%	67%	73%	*91%*
T-Cu	Copper	*reference*	*34%*	42%	53%	56%	65%	*32%*	41%	53%	56%	64%	*46%*
T-P	Phosphorus	*reference*	*33%*	41%	53%	56%	64%	*32%*	40%	52%	55%	63%	*46%*
T-Pb	Lead	*reference*	*46%*	52%	59%	61%	67%	*42%*	47%	55%	58%	64%	*81%*
T-Zn	Zinc	*reference*	*31%*	39%	49%	52%	60%	*29%*	37%	47%	50%	58%	*45%*

Notes:

WQ Inlet Parameters

Cleaning Intervals	Every 180 days *(Twice per year)*
Inlet Density	2.9 per acres *(one per 15,000 paved sq. ft.)*
Sump Storage Depth	3 ft.
Normal Sump Area	3.33 sq. ft.
Normal Sump Volume	10.0 cu. ft.
Enlarged Sump Area	12.6 sq. ft.
Enlarged Sump Volume	37.8 cu. ft.

Water Quality Vault Parameters

Vault Contributing Area	10 Acres (paved)
Vault Area	10,000 sq. ft.
Vault Depth	5 ft
Vault Volume	50,000 cu. ft.

Note - vault performance included the 100% runoff from the Roofs area.

Study Area Composition

Acres	Description
10.00	Total Study Area
5.00	A - Alleys - Pier 5 Trailer and Container Alleys, Roadways, and Parking Lots
0.50	C - Containers - Pier 5 Container Storage Areas (50% Sweepable & Washable)
0.69	T - Trailers - Pier 5 Trailer Parking & Storage Areas (8% Sweepable, 21.8% Washable)
0.50	X - Pier 5 Unsweepable Fringe between Ends of Trailers
3.31	Roofs - Contributed 100% Runoff but no pollutant loads

Damp condition sweeping parameters use values observed for tandem sweeping (standard flush / mechanical followed by vaccuum unit).

Table 16

ANNUAL ALTERNATIVE POLLUTANT CONCENTRTIONS - DAMP SWEEPING

Seattle Pier 5, Lbs per year

Pollutant Washoff		Baseline	Normal WQ Inlets					Enlarged WQ Inlets					WQ Vault
Average Annual Values (Volume-weighted Event Mean Conc'c)		*No Trapping No Sweeping*	*Sweeping Interval (days)* 30	15	7	4	3	*Sweeping Interval (days)* 30	15	7	4	3	*With Enlarged Inlets No Sweeping*
Annual Dissolved Average (mg/l)													
D-Cu	Copper	0.032	0.028	0.024	0.019	0.018	0.015	0.028	0.024	0.019	0.018	0.015	0.032
D-P	Phosphorus	0.092	0.080	0.070	0.057	0.053	0.043	0.080	0.070	0.057	0.053	0.043	0.092
D-Pb	Lead	0.002	0.002	0.002	0.001	0.001	0.001	0.002	0.002	0.001	0.001	0.001	0.002
D-Zn	Zinc	0.147	0.130	0.116	0.096	0.091	0.076	0.130	0.116	0.096	0.091	0.076	0.147
Annual Suspended Average (mg/l)													
TSS		130.7	60.3	53.0	43.1	40.4	33.1	63.9	56.2	45.7	42.8	35.2	11.5
S-Cu	Copper	0.032	0.014	0.013	0.010	0.010	0.008	0.015	0.013	0.011	0.010	0.008	0.003
S-P	Phosphorus	0.092	0.042	0.037	0.030	0.028	0.023	0.045	0.039	0.032	0.030	0.025	0.008
S-Pb	Lead	0.018	0.009	0.008	0.007	0.006	0.005	0.010	0.009	0.007	0.007	0.006	0.002
S-Zn	Zinc	0.147	0.071	0.063	0.053	0.050	0.042	0.077	0.069	0.058	0.055	0.046	0.014
Annual TOTAL Average (mg/l)													
TSS		130.7	60.3	53.0	43.1	40.4	33.1	63.9	56.2	45.7	42.8	35.2	11.5
T-Cu	Copper	0.064	0.042	0.037	0.030	0.028	0.022	0.043	0.037	0.030	0.028	0.023	0.035
T-P	Phosphorus	0.184	0.123	0.108	0.087	0.081	0.066	0.125	0.110	0.089	0.083	0.068	0.100
T-Pb	Lead	0.020	0.011	0.010	0.008	0.008	0.007	0.011	0.010	0.009	0.008	0.007	0.004
T-Zn	Zinc	0.293	0.202	0.179	0.150	0.141	0.119	0.208	0.185	0.154	0.145	0.122	0.160
Annual Percent Removals													
TSS		*reference*	*54%*	59%	67%	69%	75%	*51%*	57%	65%	67%	73%	*91%*
T-Cu	Copper	*reference*	*34%*	42%	53%	56%	65%	*32%*	41%	53%	56%	64%	*46%*
T-P	Phosphorus	*reference*	*33%*	41%	53%	56%	64%	*32%*	40%	52%	55%	63%	*46%*
T-Pb	Lead	*reference*	*46%*	52%	59%	61%	67%	*42%*	47%	55%	58%	64%	*81%*
T-Zn	Zinc	*reference*	*31%*	39%	49%	52%	60%	*29%*	37%	47%	50%	58%	*45%*

Notes:

WQ Inlet Parameters

Cleaning Intervals	Every 180 days *(Twice per year)*
Inlet Density	2.9 per acres *(one per 15,000 paved sq. ft.)*
Sump Storage Depth	3 ft.
Normal Sump Area	3.33 sq. ft.
Normal Sump Volume	10.0 cu. ft.
Enlarged Sump Area	12.6 sq. ft.
Enlarged Sump Volume	37.8 cu. ft.

Water Quality Vault Parameters

Vault Contributing Area	10 Acres (paved)
Vault Area	10,000 sq. ft.
Vault Depth	5 ft
Vault Volume	50,000 cu. ft.

Note - vault performance included the 100% runoff from the Roofs area.

Study Area Composition

Acres	Description
10.00	Total Study Area
5.00	A - Alleys - Pier 5 Trailer and Container Alleys, Roadways, and Parking Lots
0.50	C - Containers - Pier 5 Container Storage Areas (50% Sweepable & Washable)
0.69	T - Trailers - Pier 5 Trailer Parking & Storage Areas (8% Sweepable, 21.8% Washable)
0.50	X - Pier 5 Unsweepable Fringe between Ends of Trailers
3.31	Roofs - Contributed 100% Runoff but no pollutant loads

Damp condition sweeping parameters use values observed for tandem sweeping (standard flush / mechanical followed by vaccuum unit).

Total Runoff volume is 54.0 Acre-feet (mg/l = 0.3677 * lbs/acreft)

Table 17

EXPECTED RANGE OF ANNUAL POLLUTANT LOAD REDUCTIONS ASSOCIATED WITH VARIOUS SWEEPING FREQUENCIES

	Twice Weekly Sweeping	Weekly Sweeping	Biweekly Sweeping	Wet Vaults
TSS	45 - 70%	45 - 65%	40 - 60%	75 - 90%
Total Phosphorus	35 - 60%	30 - 55%	20 - 40%	35 - 45%
Total Lead	40 - 60%	35 - 60%	30 - 50%	65 - 80%
Total Zinc	30 - 55%	25 - 50%	20 - 40%	35 - 45%
Total Copper	35 - 60%	30 - 55%	20 - 40%	35 - 45%

Normally sized catchbasins are cleaned once during the year.

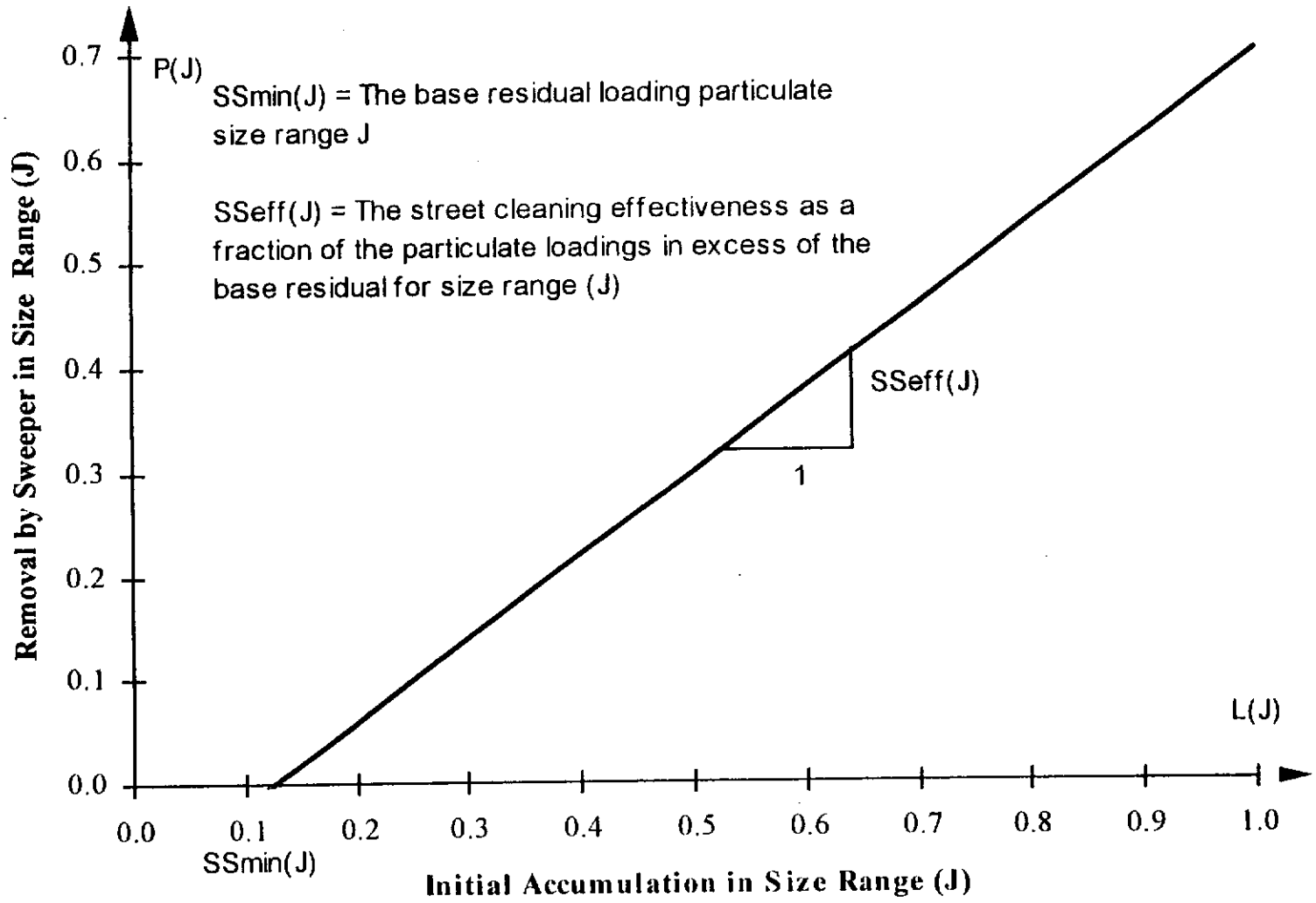
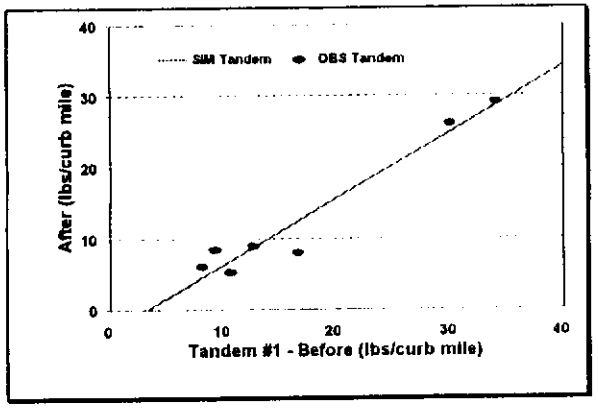


Figure 1
Street Sweeping Model Component of SIMPTM

Size Group 1

< 63 microns

SSeff = 93%
SSmin = 2.0
Lbs/Paved Acre

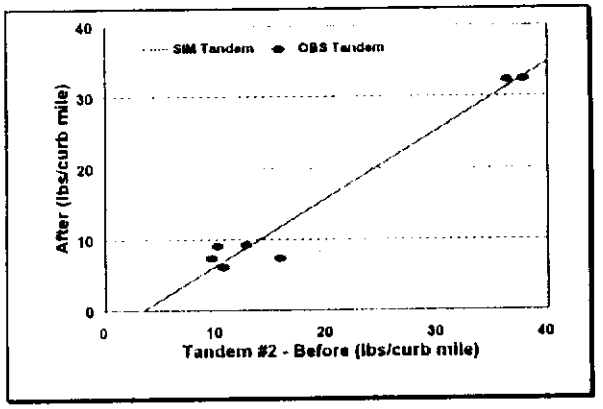


Size Group 2

< 125 microns

R2 = 0.962

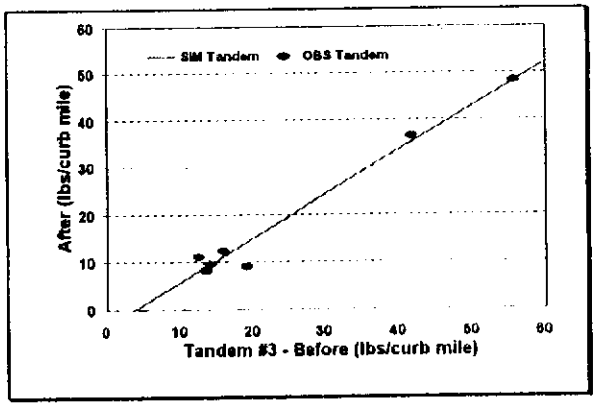
SSeff = 95%
SSmin = 2.0
Lbs/Paved Acre



Size Group 3

< 250 microns

R2 = 0.975
SSeff = 93%
SSmin = 2.3
Lbs/Paved Acre



Size Group 4

< 600 microns

R2 = 0.988
SSeff = 89%
SSmin = 2.3
Lbs/Paved Acre

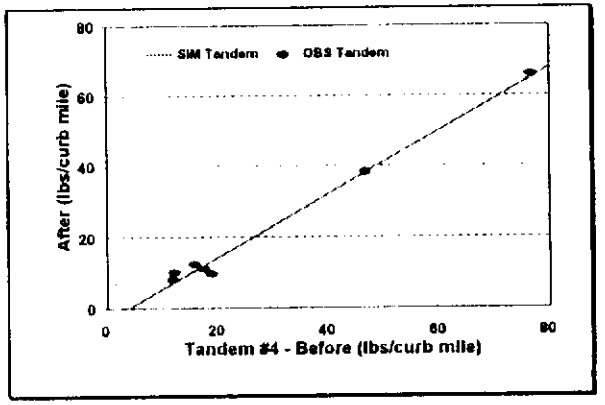
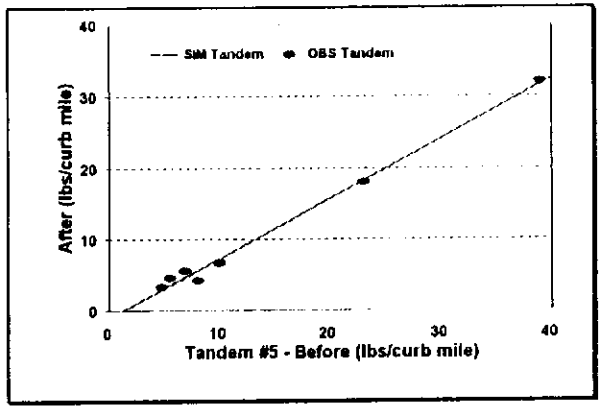


Figure 2
Tandem Street Sweeping Model in SIMPTM, Size Groups 1-4

Size Group 5

< 1000 microns

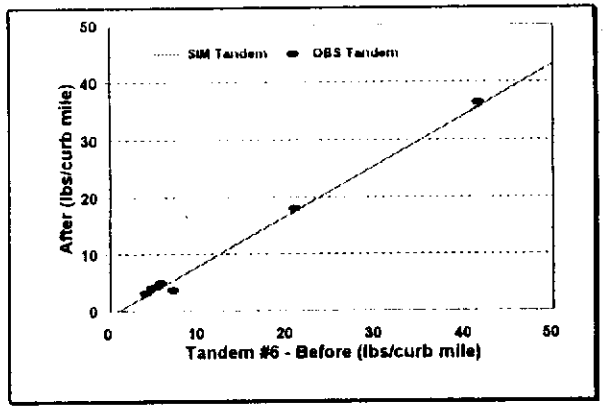
R2 = 0.992
SSeff = 84%
SSmin = 0.8
Lbs/Paved Acre



Size Group 6

< 2000 microns

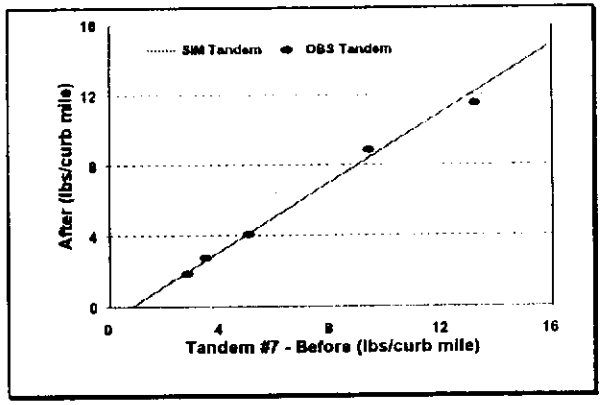
R2 = 0.997
SSeff = 88%
SSmin = 0.6
Lbs/Paved Acre



Size Group 7

< 6370 microns

R2 = 0.999
SSeff = 98%
SSmin = 0.5
Lbs/Paved Acre



Size Group 8

> 6370 microns

R2 = 0.995
SSeff = 87%
SSmin = 0.0
Lbs/Paved Acre

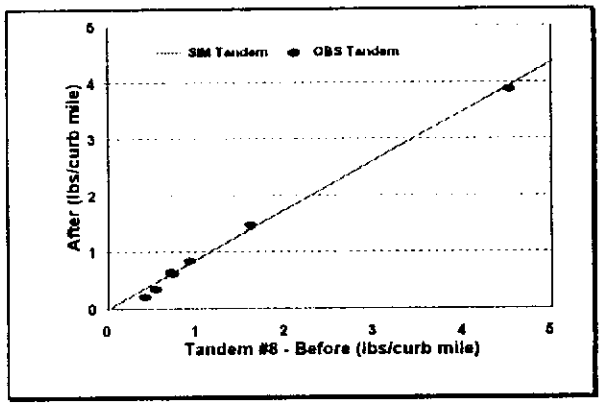


Figure 2 (continued)
Tandem Street Sweeping Model in SIMPTM, Size Groups 5-8

Figure 3

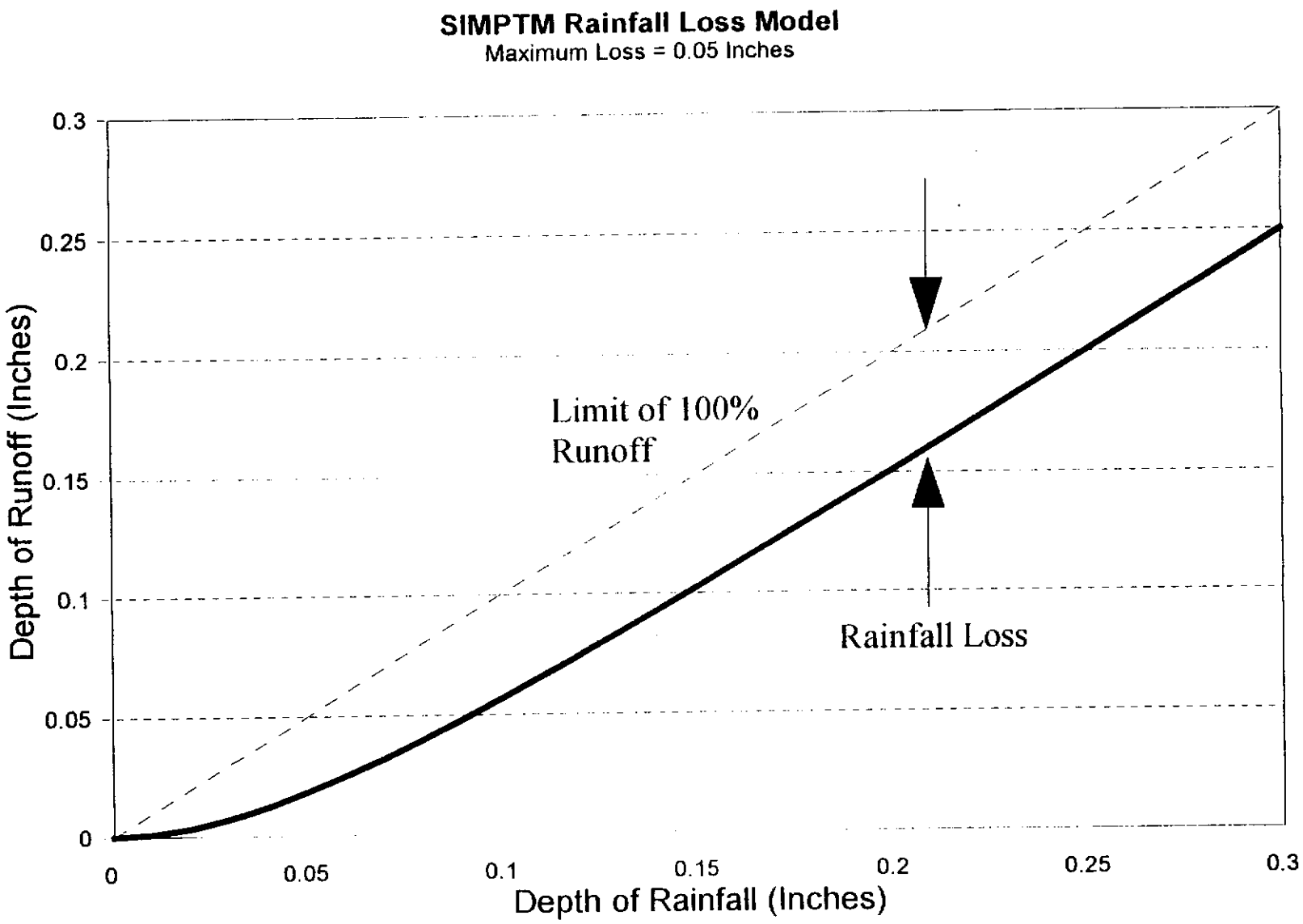

SIMPTM Rainfall Loss Model
Maximum Loss = 0.05 Inches
Depth of Runoff (Inches)
0.3
0.25
0.2
0.15
0.1
0.05
0
Limit of 100% Runoff
Rainfall Loss
0
0.05
0.1
0.15
0.2
0.25
0.3
Depth of Rainfall (Inches)

Figure 4
Dry Weather Accumulation Model

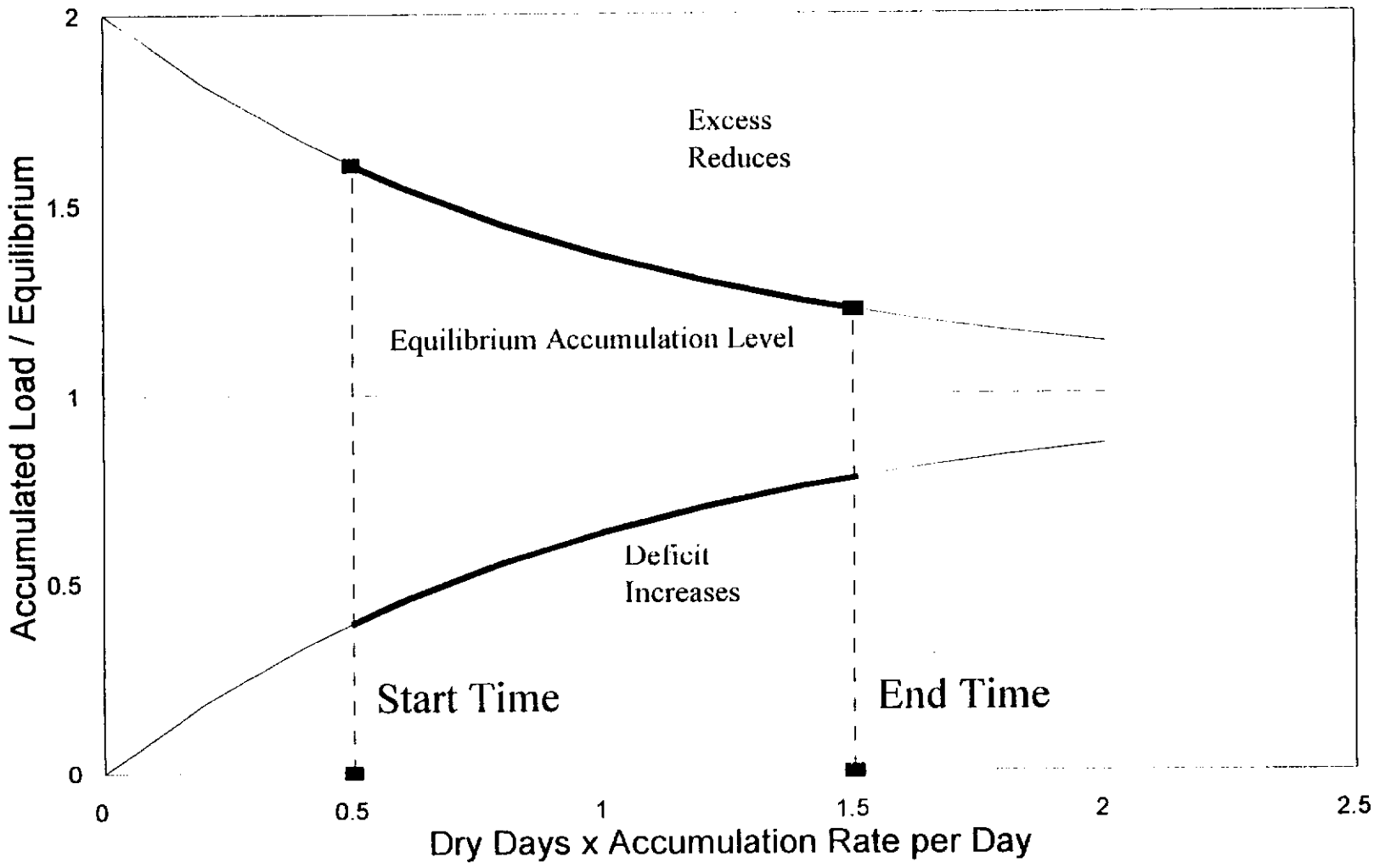


Figure 5
Accumulation Rate - Container Alley CA

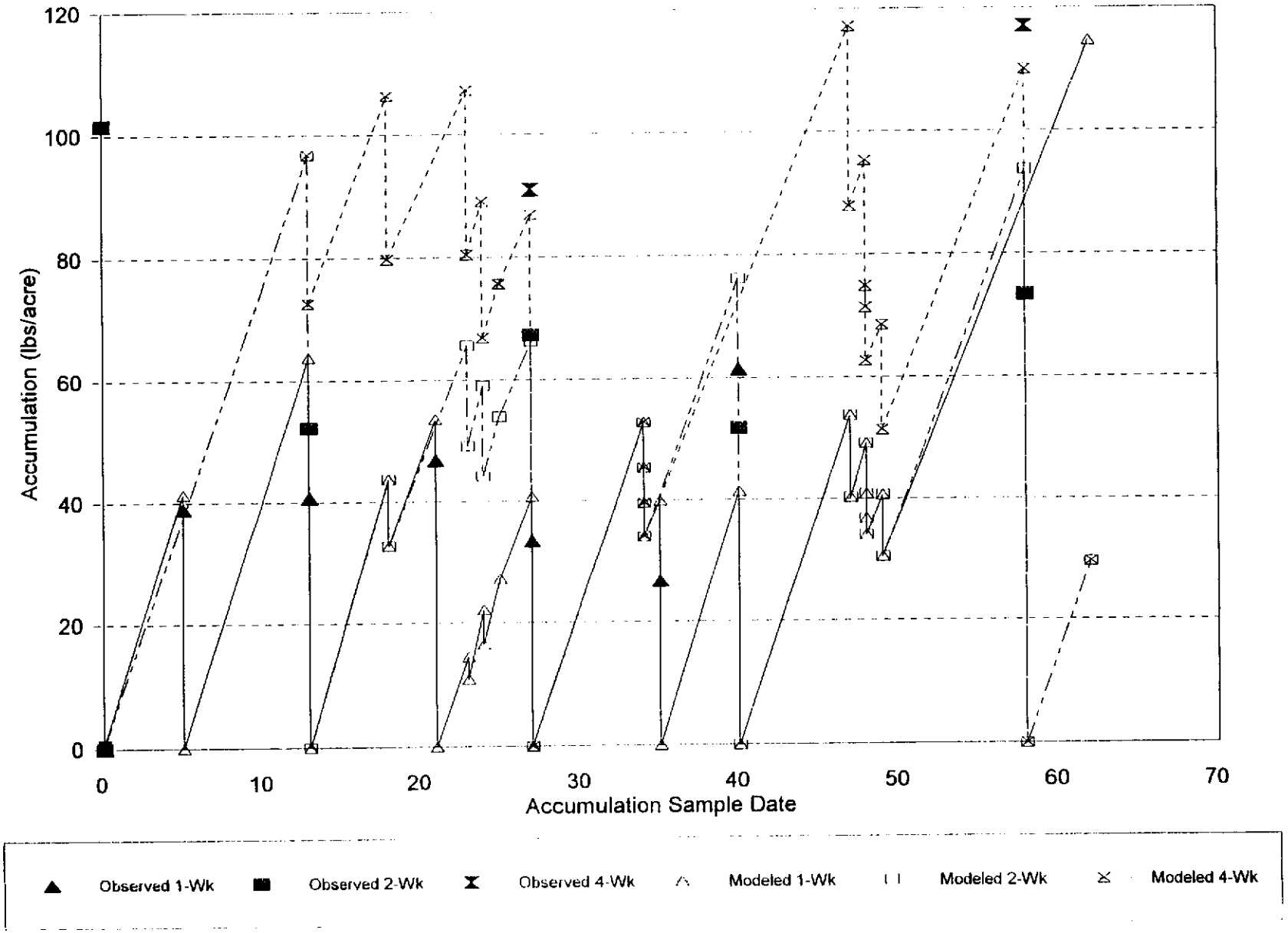


Figure 6
Accumulation Rate - Trailer Alley TA

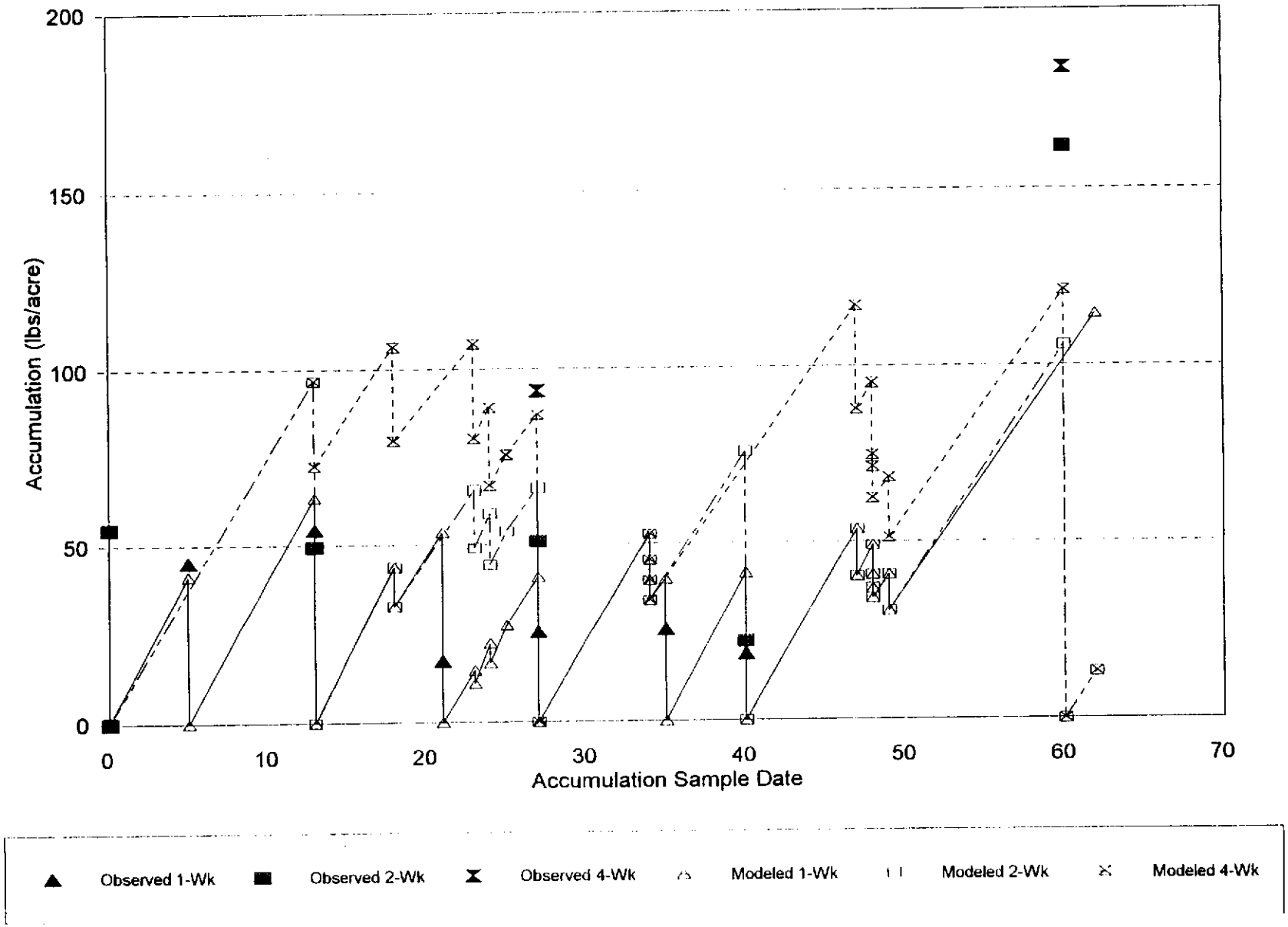


Figure 7
Accumulation Rate - Trailer Parking TP

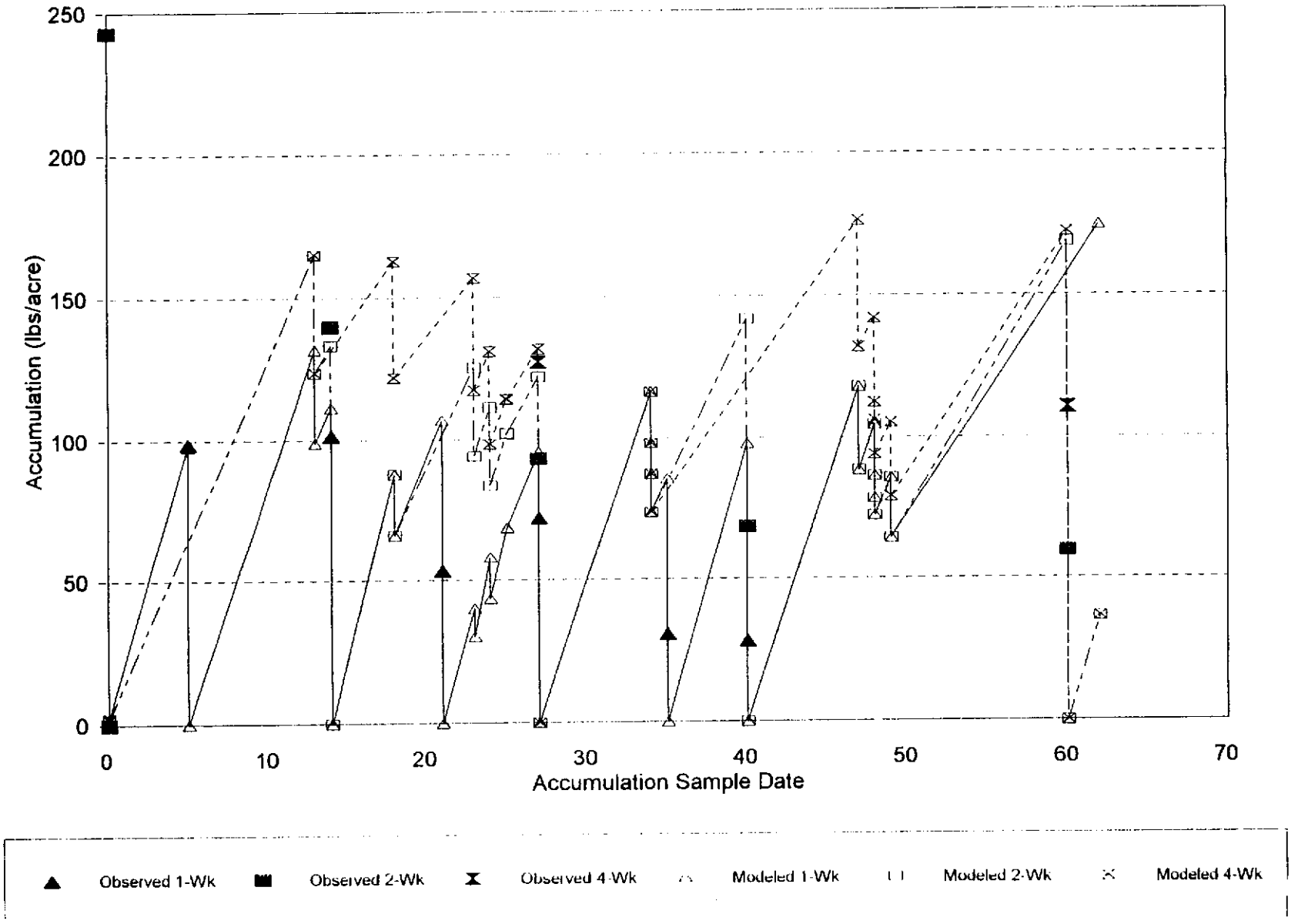


Figure 8

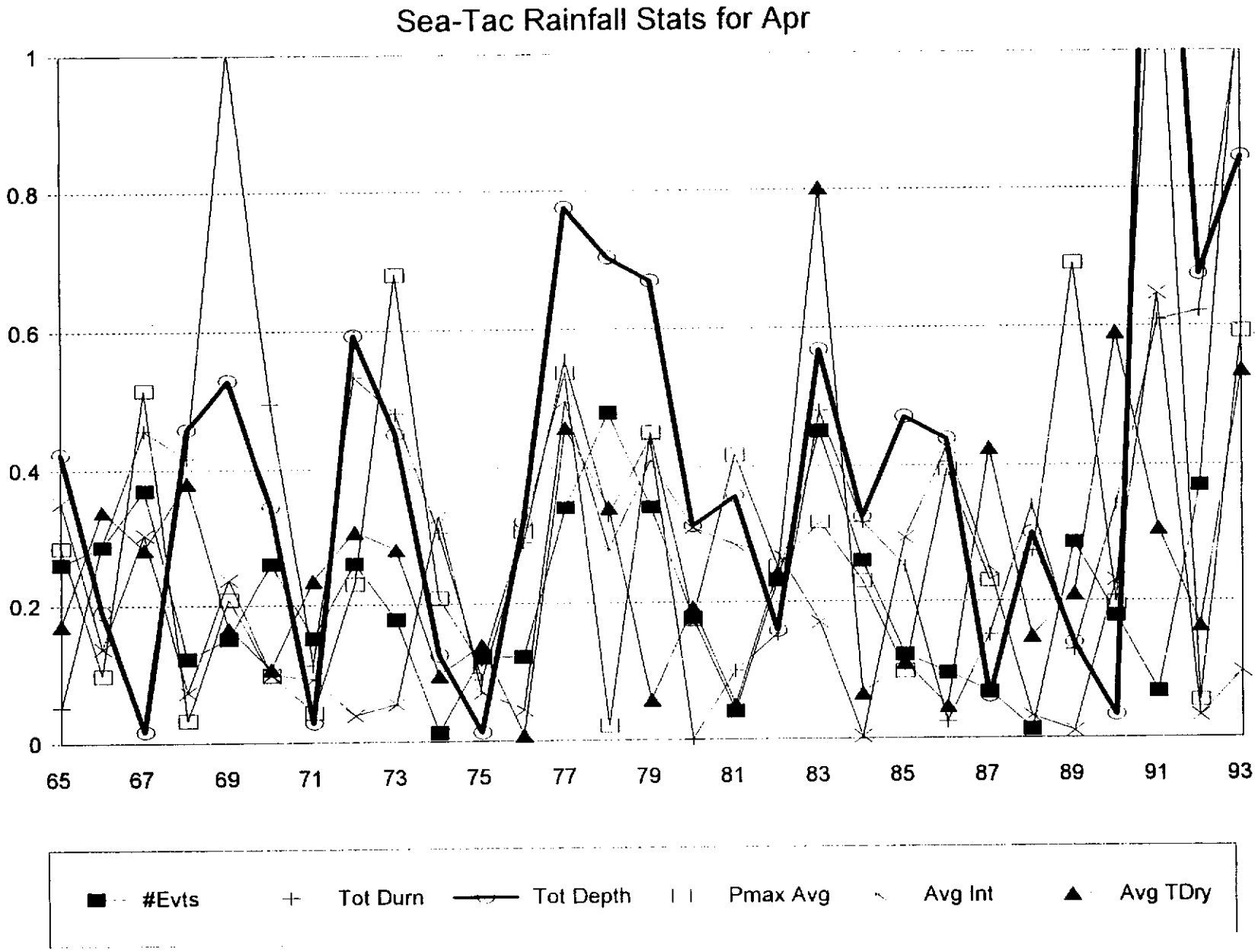
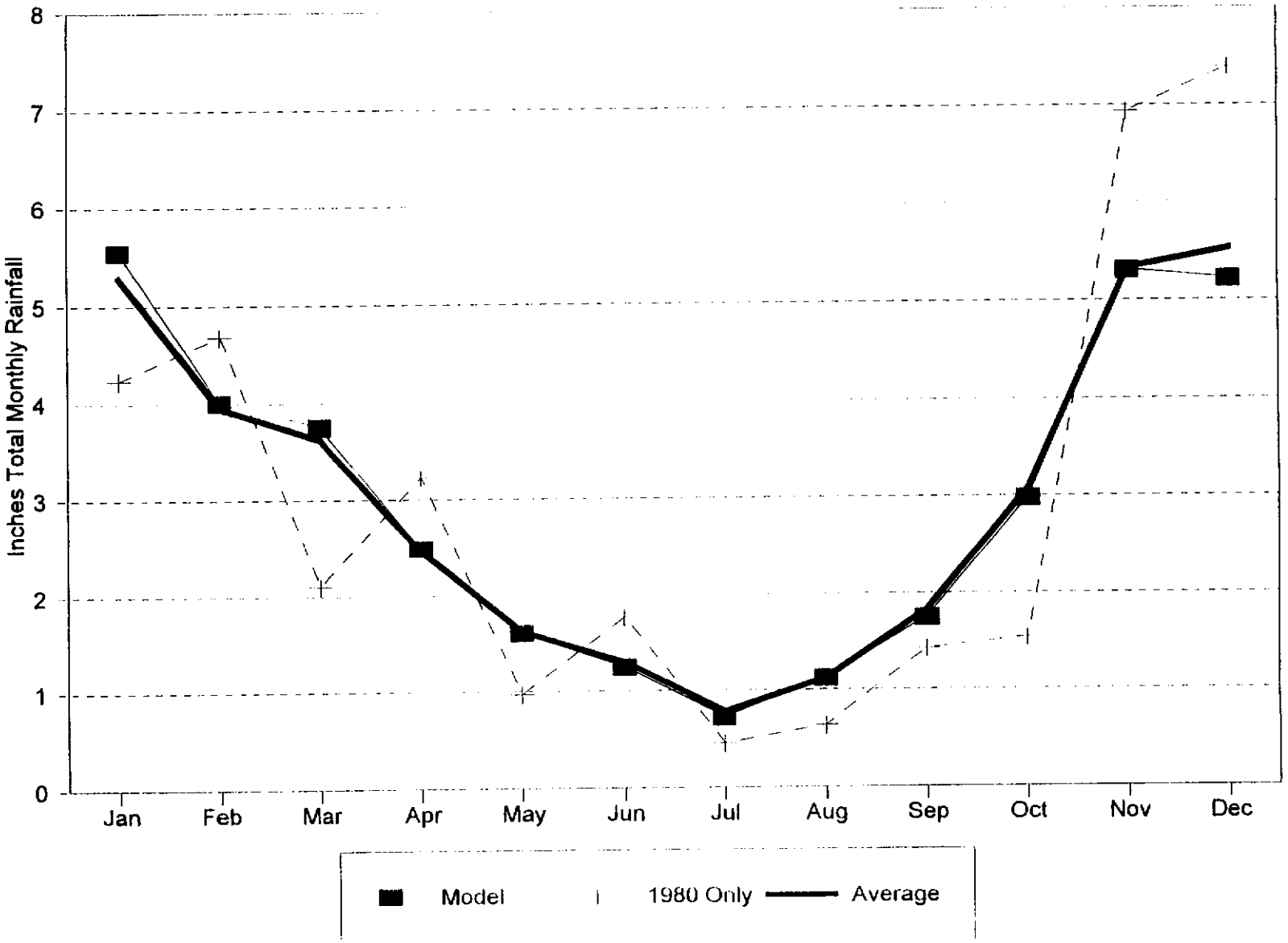

Sea-Tac Rainfall Stats for Apr
1
0.8
0.6
0.4
0.2
0
65
67
69
71
73
75
77
79
81
83
85
87
89
91
93
#Evts
Tot Durn
Tot Depth
Pmax Avg
Avg Int
Avg TDry

Figure 9

Model Year Rainfall Comparison

Sea-Tac Airport NOAA Hourly Data



ADDENDUM #1

PORT OF SEATTLE

STORMWATER TREATMENT BMP EVALUATION

Subsequent to the above referenced report submitted in March 1997, several formatting errors were discovered in a few of the tables and a clarification of one statement was requested.

It was felt that some additional clarification was desired for the statement that appeared both on page iv of the Executive Summary (first paragraph after item 8) and page 14 of the main body of the report. The statement of concern is:

> "With regard to sweeping frequency, the model suggests there may be significant benefit in sweeping weekly as opposed to biweekly. However, there may be little incremental benefit of sweeping twice weekly in comparison to weekly."

If one looks at the "Annual Percent Removals" simulated for various sweeping intervals (i.e. Tables 13 - 16), it appears that the increase in removal when one goes from a sweeping program of every 15 days to every 7 days is very similar to the increase in removal simulated from 7 days to 3 days. This implies that the above statement is not correct. Therefore, some clarification on this important issue is needed.

What should have been written was the belief by the authors that weekly sweeping appears to be the optimum sweeping program. The dramatic reduction from sweeping every 4 days to every 3 days shown in these tables (i.e. 13 - 16) does not appear to be realistic and is probably the result of a complex model interaction that actually allows more sweeping events to occur during the "3 day" simulation because of significant rainfall interference that may be occurring during the "4 day" simulation. It is probably more realistic to average the 4 day and 3 day sweeping removals and assign them to a twice weekly sweeping program. If this is done, one can see that the increase in removal when one goes from every 15 days to every 7 days (i.e. 108% increase in effort) is much greater than the increase in removal when one goes from every 7 days to every 3.5 days (i.e. 100% increase in effort). Thus, the optimum level of effort appears to be weekly sweeping since the incremental benefit of doubling the effort and cost from 7 to 3.5 days is considerably less than the 108% increase in effort and cost from 15 to 7 days. This is the logic behind the first study conclusion documented on page iii of the Executive Summary and page 13 of the main body of the report and the statement that appears earlier in quotes.